Uncertainty visualization

Jacques Gautier

LaSTIG, GeoVIS Team, Geodata Paris, IGN, DCAIG

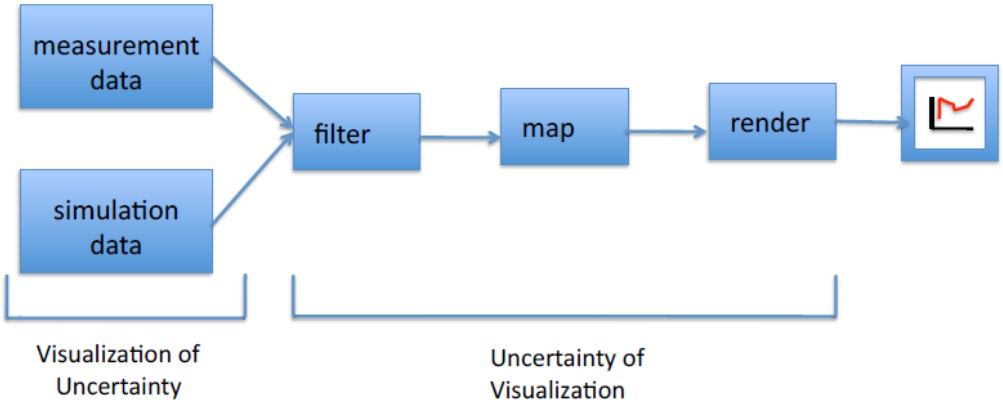
Representing data uncertainty

Data visualization is a powerful tool for supporting decision-making, but the represented data may be subject to uncertainty coming from various origins:

Representing data uncertainty

Data visualization is a powerful tool for supporting decision-making, but the represented data may be subject to uncertainty coming from various origins:

- ***measurement****: manipulation or calibration errors, sensors precision, sampling,*
- ***modeling*** *:model approximation and sensitivity, …*
- ***visualization****: data selection and aggregation, chosen approach and visual channels*



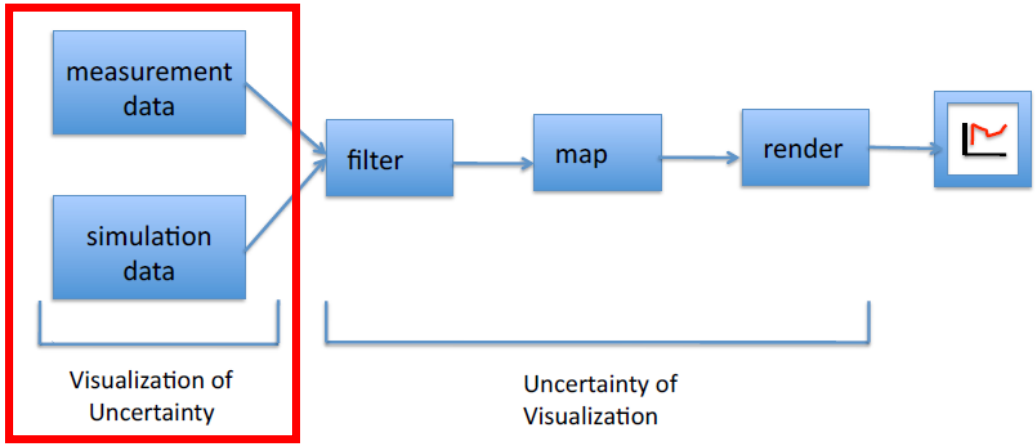
Robert B. Haber and David A. McNabb. Visualization idioms: A conceptual model for scientific visualization systems. In B. Shriver, G.M. Nielson, and L.J. Rosenblum, editors, Visualization in Scientific Computing, pages 74–93. IEEE, 1990.

BRODLIE, Ken, ALLENDES OSORIO, Rodolfo, et LOPES, Adriano. A review of uncertainty in data visualization. *Expanding the frontiers of visual analytics and visualization*, 2012, p. 81-109.

Representing data uncertainty

Data visualization is a powerful tool for supporting decision-making, but the represented data may be subject to uncertainty coming from various origins:

- ***measurement****: manipulation or calibration errors, sensors precision, sampling,*
- ***modeling*** *:model approximation and sensitivity, …*
- ***visualization****: data selection and aggregation, chosen approach and visual channels*



Robert B. Haber and David A. McNabb. Visualization idioms: A conceptual model for scientific visualization systems. In B. Shriver, G.M. Nielson, and L.J. Rosenblum, editors, Visualization in Scientific Computing, pages 74–93. IEEE, 1990.

BRODLIE, Ken, ALLENDES OSORIO, Rodolfo, et LOPES, Adriano. A review of uncertainty in data visualization. *Expanding the frontiers of visual analytics and visualization*, 2012, p. 81-109.

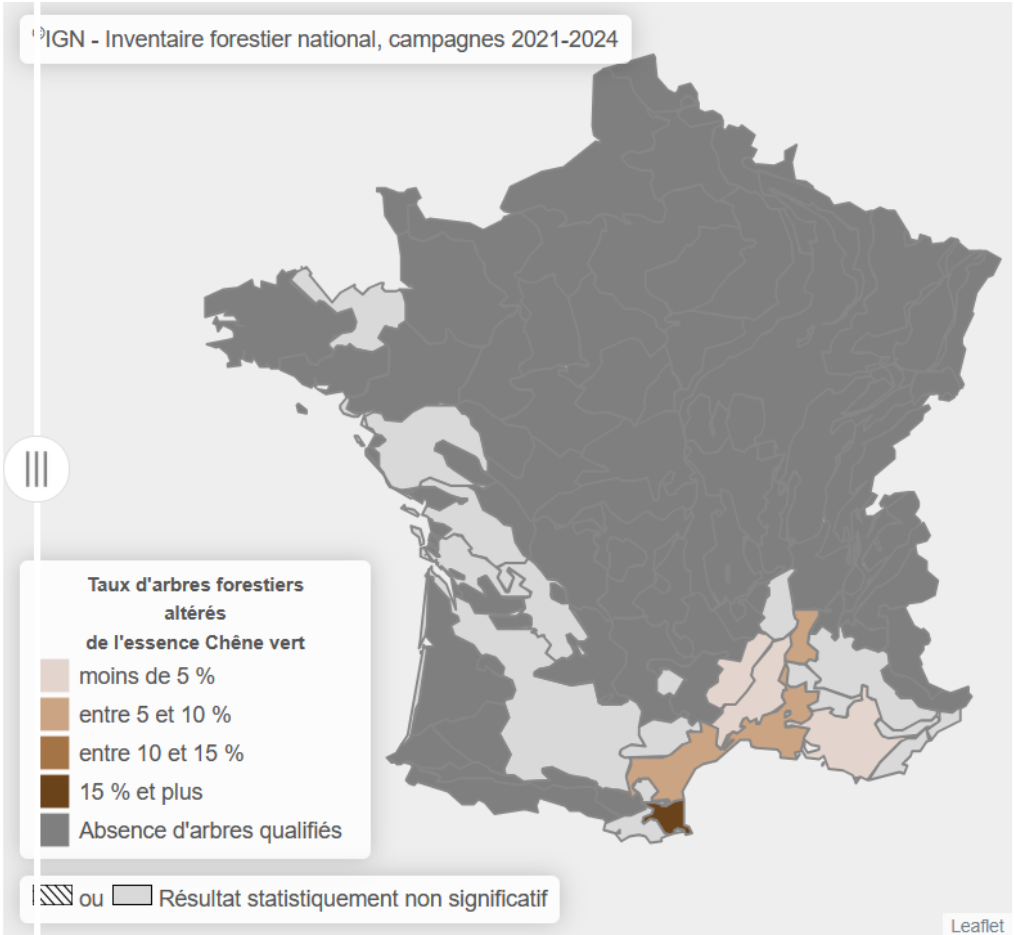
Visualizing data uncertainty

Most visualization approaches show data as being accurate

BRODLIE, Ken, ALLENDES OSORIO, Rodolfo, et LOPES, Adriano. A review of uncertainty in data visualization. Expanding the frontiers of visual analytics and visualization, 2012, p. 81-109.

What should be done when data is tainted by uncertainty?

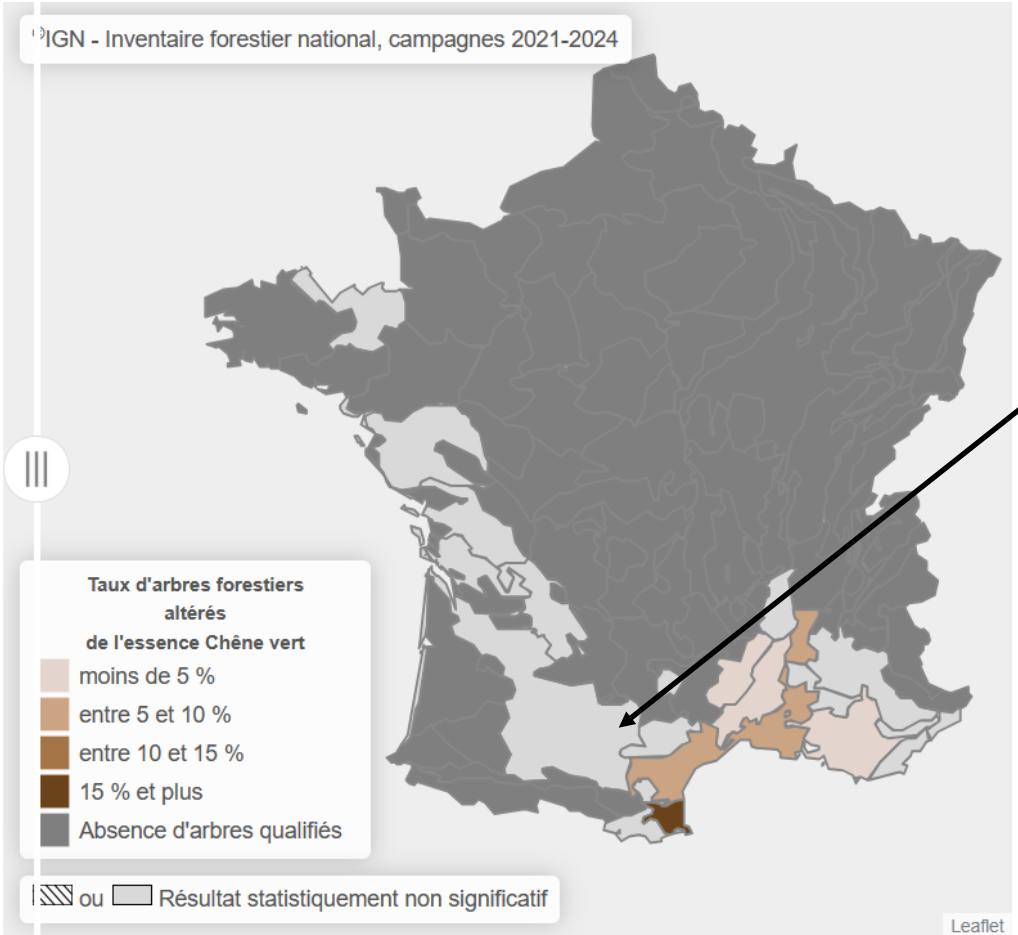
Visualizing data uncertainty



Not visualizing it ?

https://inventif.ign.fr

Visualizing data uncertainty

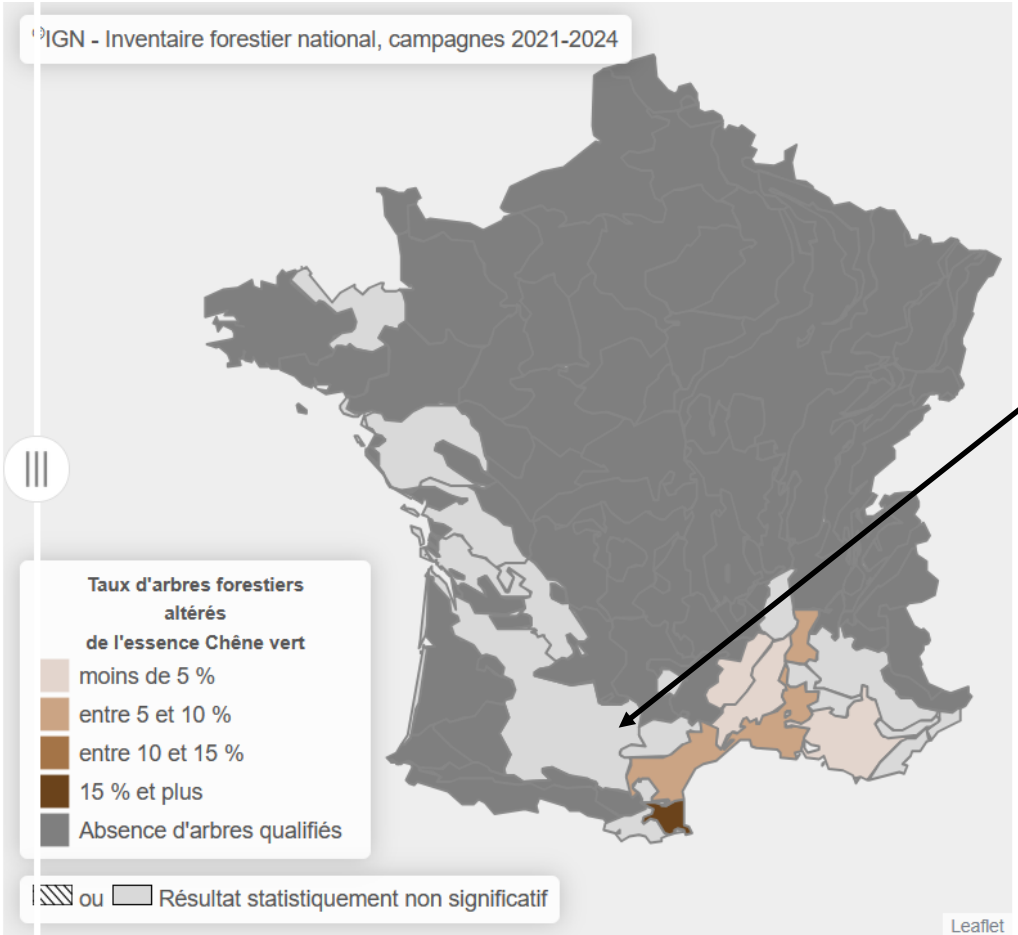


Not visualizing it ?

What will the user understand?

https://inventif.ign.fr

Visualizing data uncertainty



Not visualizing it ?

What will the user understand?

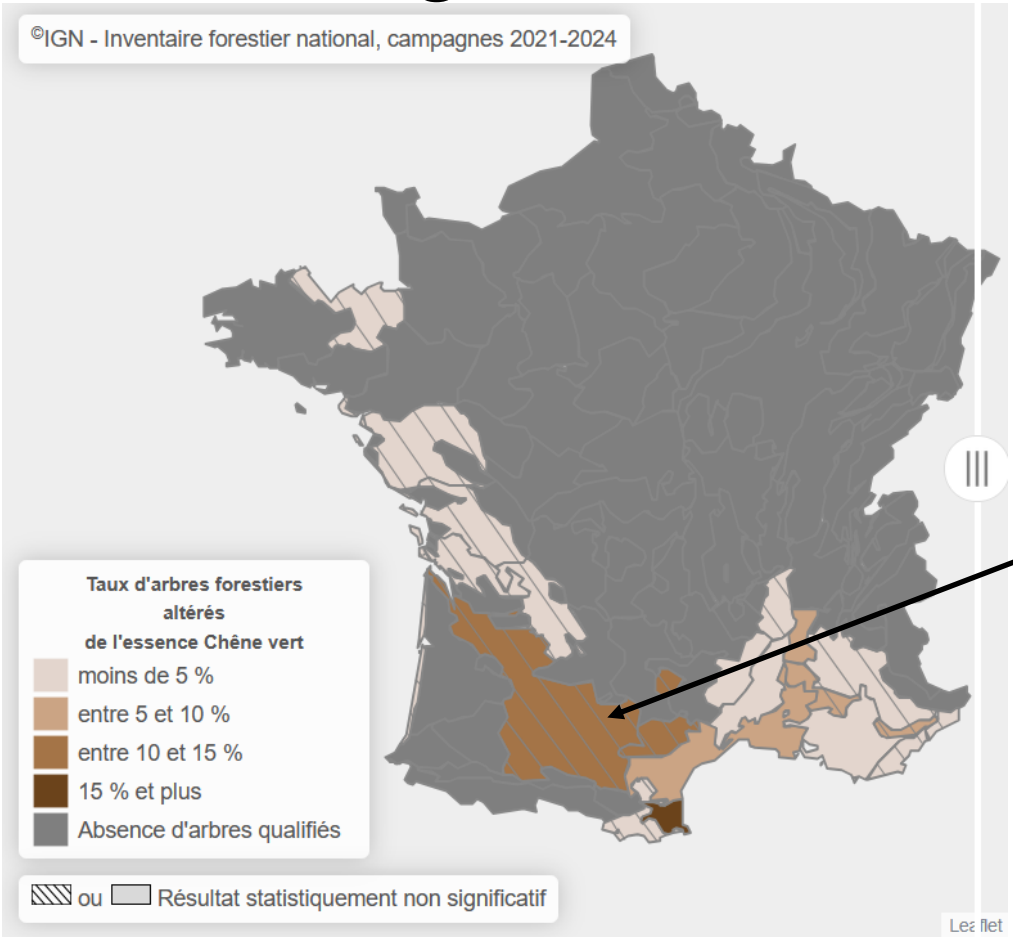
« There is data with too much uncertainty » ?

or

« There is no data at all »

https://inventif.ign.fr

Visualizing data uncertainty

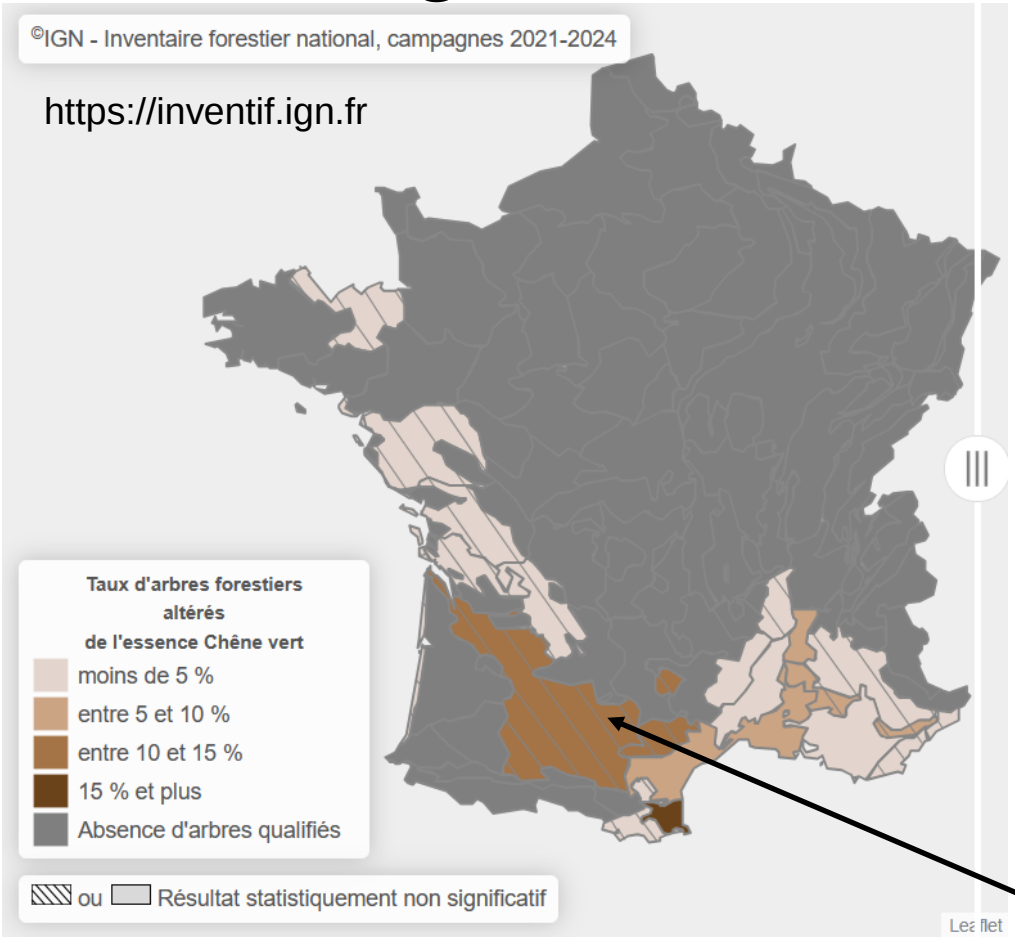


Co-visualize data and uncertainty around it

« We have data, but it is not significant »

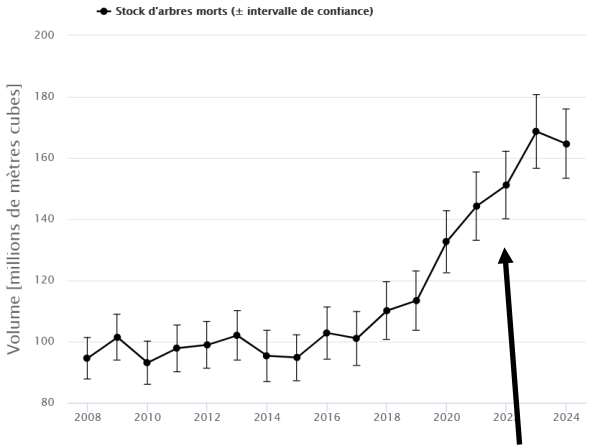
https://inventif.ign.fr

Visualizing data uncertainty



Co-visualize data and uncertainty around it

But how?



« Dead tree volume is …, + or – … »

« *The data is not significant* »

Visualizing data uncertainty

Most visualization approaches show data as being accurate

BRODLIE, Ken, ALLENDES OSORIO, Rodolfo, et LOPES, Adriano. A review of uncertainty in data visualization. Expanding the frontiers of visual analytics and visualization, 2012, p. 81-109.

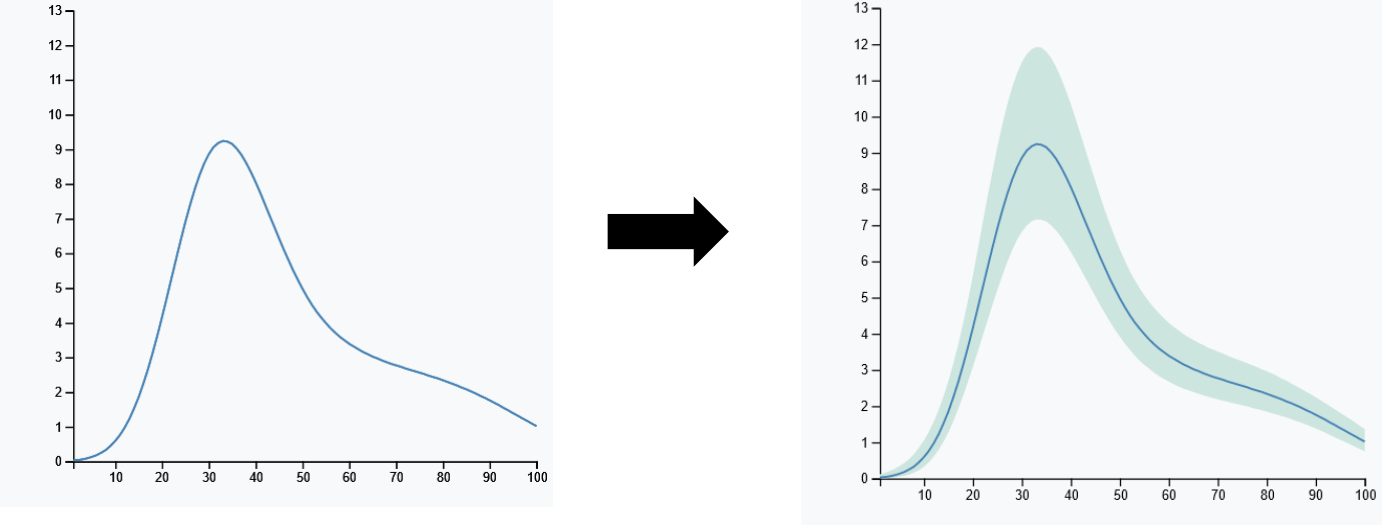
What should be done when data is tainted by uncertainty?

If we co-visualize data and data uncertainty, how to do it?

Visualizing data uncertainty

Adding an additional dimension to data

→ *Adding an extra layer of complexity to visualizations that may already be complex*



Visualizing data uncertainty

Uncertainty can be defined, and visualized, differently…
…depending on the represented data

Is it election pool results with a margin of error?

Is it a measurement considered as representative or not ?

Visualizing data uncertainty

Uncertainty can be defined, and visualized, differently…
…depending on the represented data

Is it election pool results with a margin of error?

Is it a measurement considered as representative or not ?

…depending of the targeted user

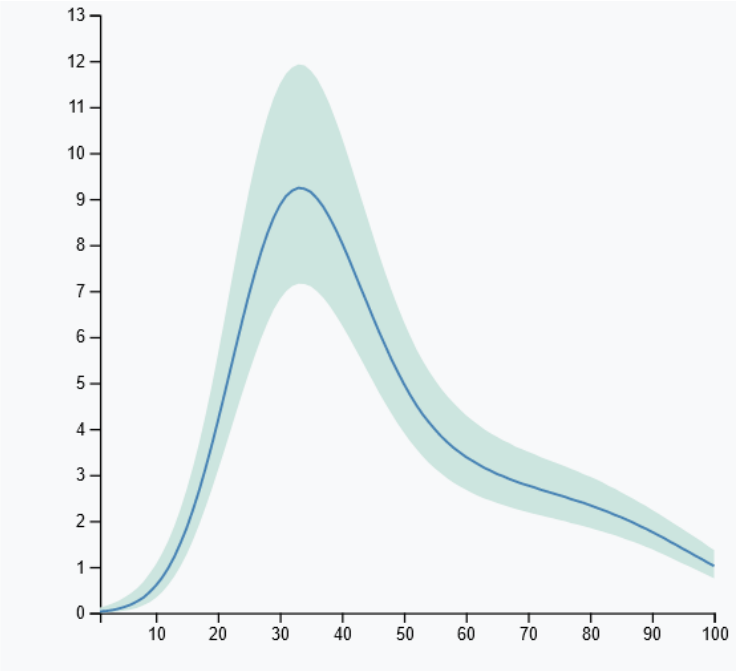
« I'm a forecaster and I want to see the entire possible extent of a phenomena and its degree of plausibility. »

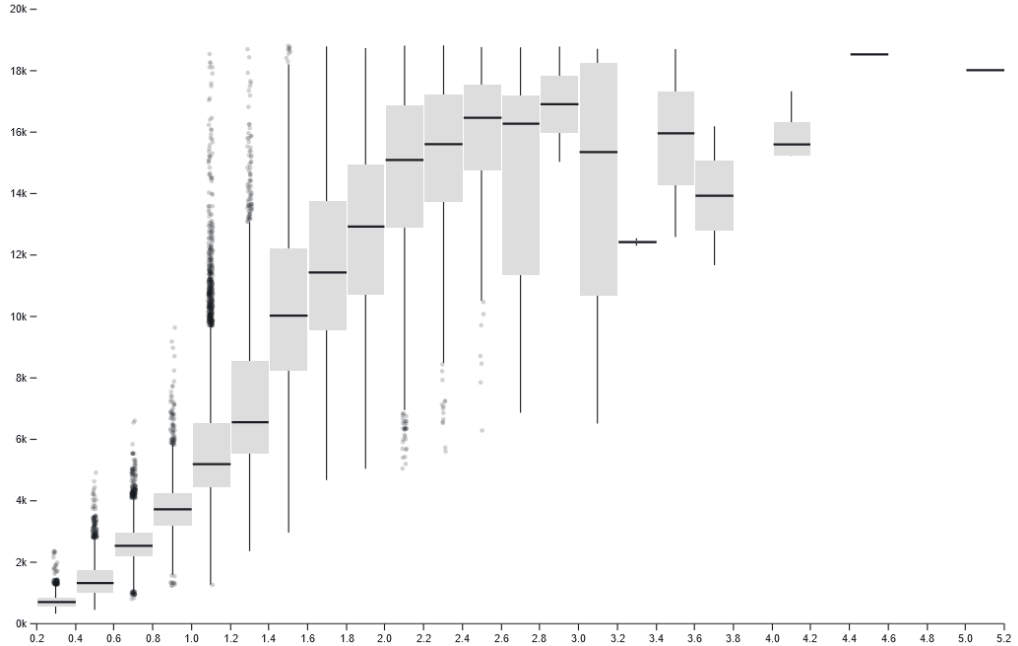
« I'm a public official, and I need an estimated potential cost overrun for the construction of a building. »

« I'm a prefect, and I need to know if, « Yes or No », I have to evacuate a neighborhood. »

Visualizing data uncertainty

Uncertainty can be defined, and visualized, differently…





Visualizing the very concept of uncertainty

Uncertainty about the dataset itself:
Estimated/simulated data
Blury spatial component

Visualizing the overall dataset as uncertain

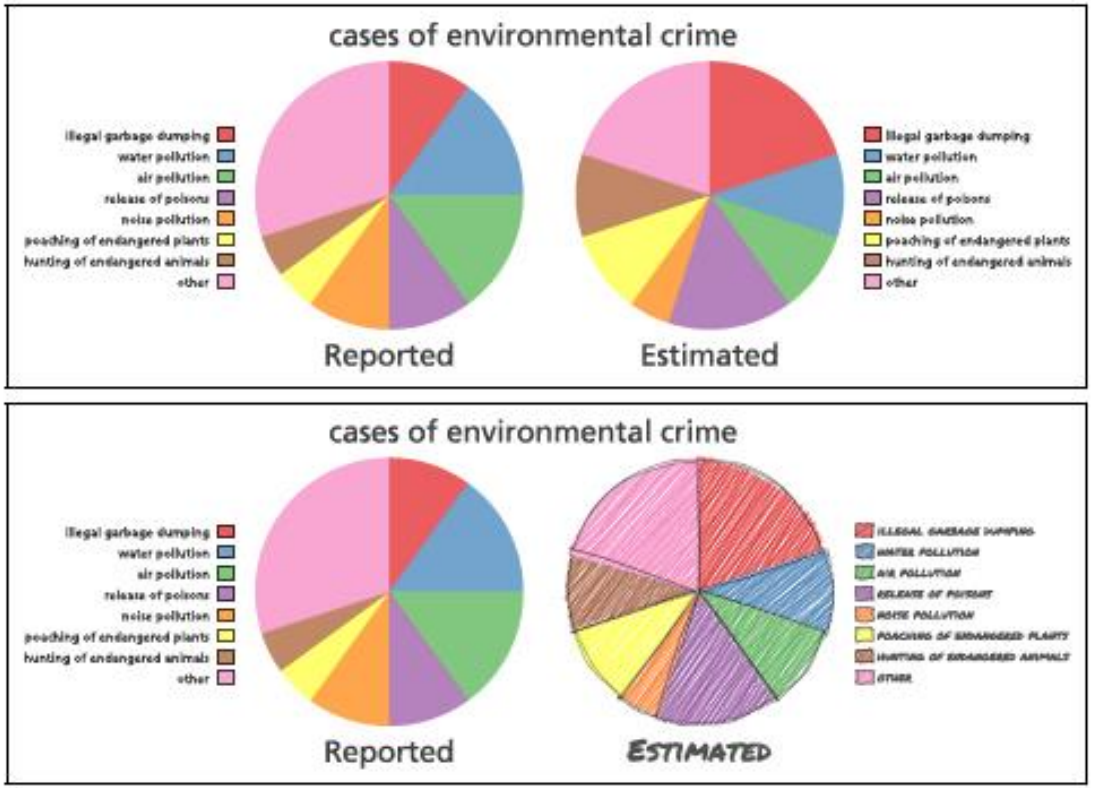


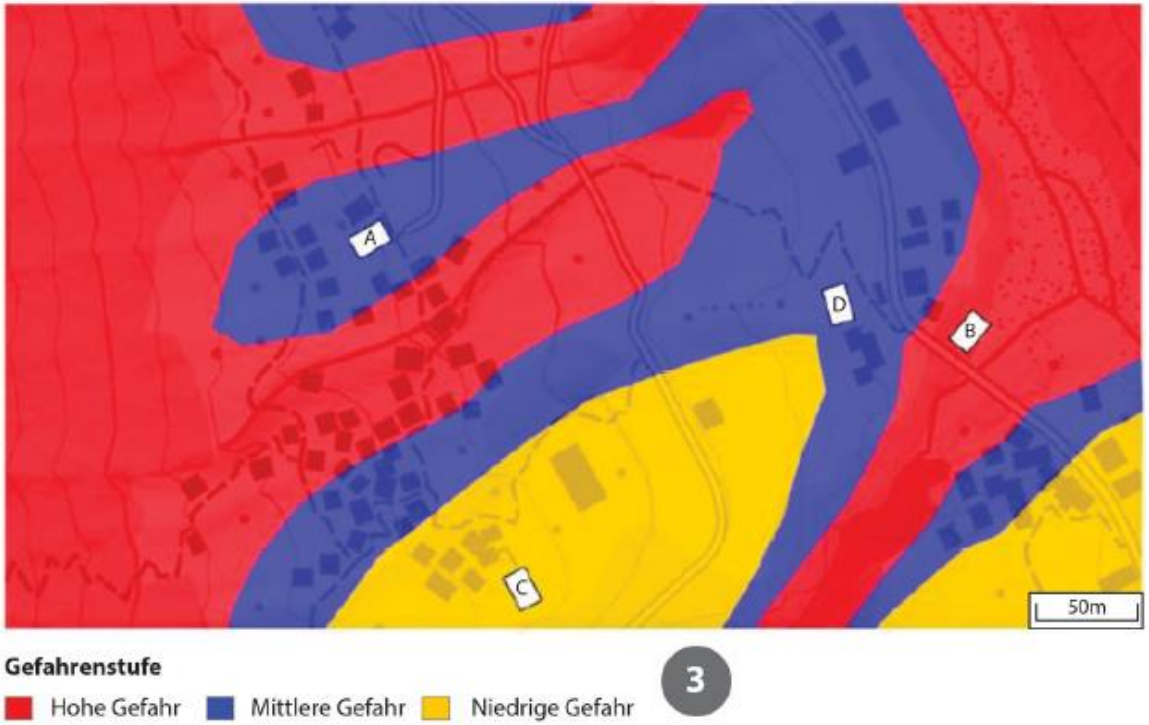
Fig. 15. Non-sketchy and hybrid visualizations for Scenario 3.

Wood, J., Isenberg, P., Isenberg, T., Dykes, J., Boukhelifa, N., & Slingsby, A. (2012). Sketchy rendering for information visualization. IEEE transactions on visualization and computer graphics, 18(12), 2749-2758.

Visualizing the very concept of uncertainty

Uncertainty about the dataset itself:
Estimated/simulated data
Blury spatial component

Visualizing the overall dataset as uncertain



KÜBLER, Isabella, RICHTER, Kai-Florian, et FABRIKANT, Sara Irina. Against all odds: multicriteria decision making with hazard prediction maps depicting uncertainty. *Annals of the American Association of Geographers*, 2020, vol. 110, no 3, p. 661-683.

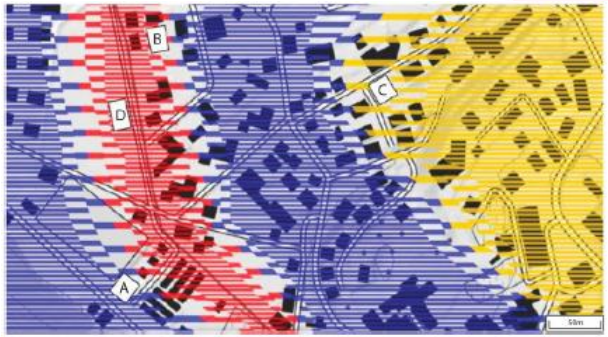
Visualizing the very concept of uncertainty

Uncertainty about the dataset itself:
Estimated/simulated data
Blury spatial component





Visualizing the overall dataset as uncertain

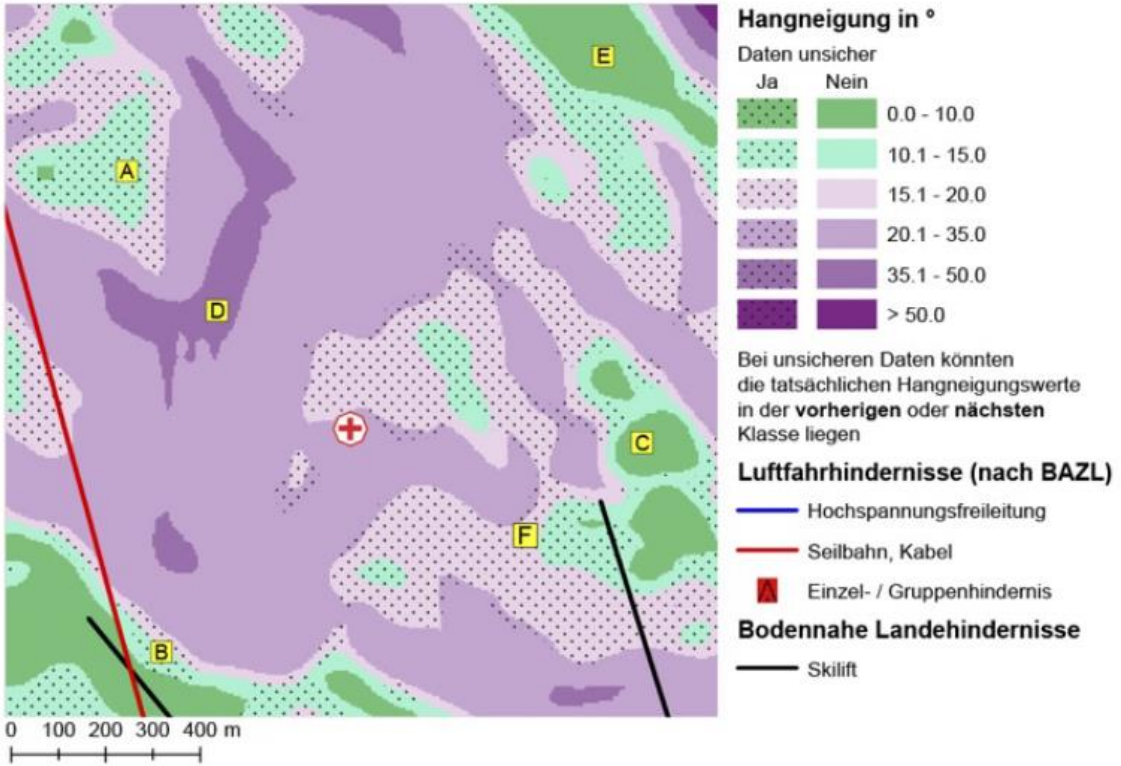


KÜBLER, Isabella, RICHTER, Kai-Florian, et FABRIKANT, Sara Irina. Against all odds: multicriteria decision making with hazard prediction maps depicting uncertainty. *Annals of the American Association of Geographers*, 2020, vol. 110, no 3, p. 661-683.

Discriminate uncertain data from the others

Uncertain/Certain values
Non significant values

Visualizing uncertain as a boolean



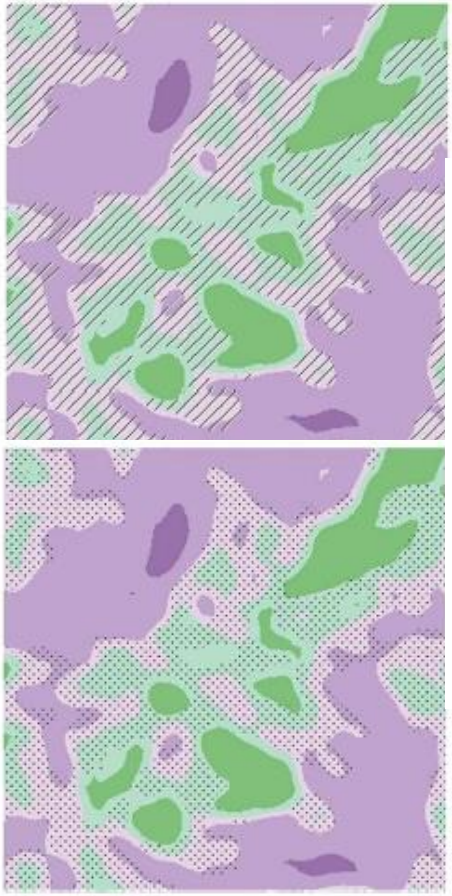
KORPORAAL, Michelle et FABRIKANT, Sara Irina. How are map-based decisions influenced by uncertainty visualization in risky and time-critical situations?. *Abstracts of the ICA*, 2019, vol. 1, p. 184.

KORPORAAL, Michelle, RUGINSKI, Ian T., et FABRIKANT, Sara Irina. Effects of uncertainty visualization on map-based decision making under time pressure. *Frontiers in Computer Science*, 2020, vol. 2, p. 32.

Discriminate uncertain data from the others

Uncertain/Certain values
Non significant values

Visualizing uncertain as a boolean



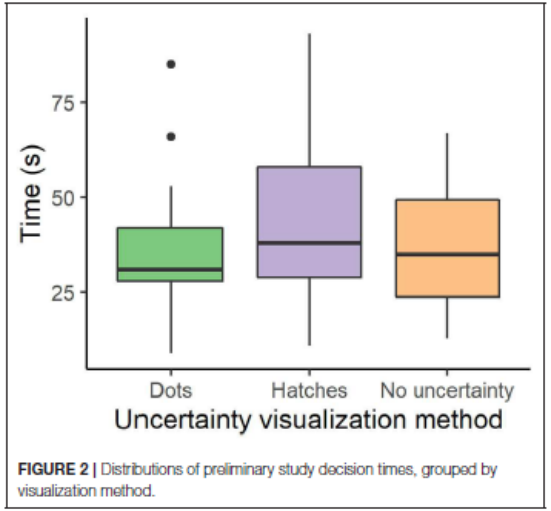


FIGURE 2 | Distributions of preliminary study decision times, grouped by visualization method.

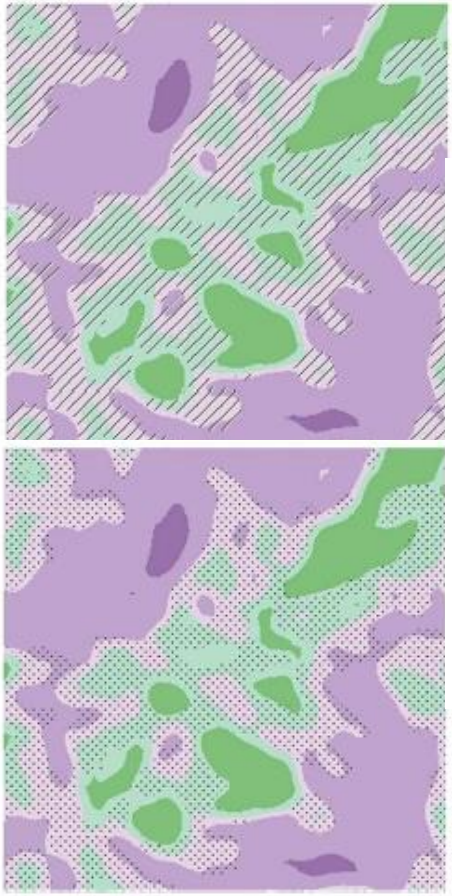
KORPORAAL, Michelle et FABRIKANT, Sara Irina. How are map-based decisions influenced by uncertainty visualization in risky and time-critical situations?. *Abstracts of the ICA*, 2019, vol. 1, p. 184.

KORPORAAL, Michelle, RUGINSKI, Ian T., et FABRIKANT, Sara Irina. Effects of uncertainty visualization on map-based decision making under time pressure. *Frontiers in Computer Science*, 2020, vol. 2, p. 32.

Discriminate uncertain data from the others

Uncertain/Certain values
Non significant values

Visualizing uncertain as a boolean



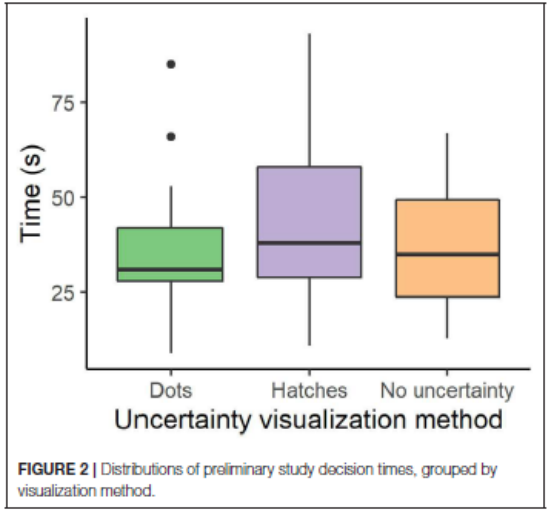


FIGURE 2 | Distributions of preliminary study decision times, grouped by visualization method.

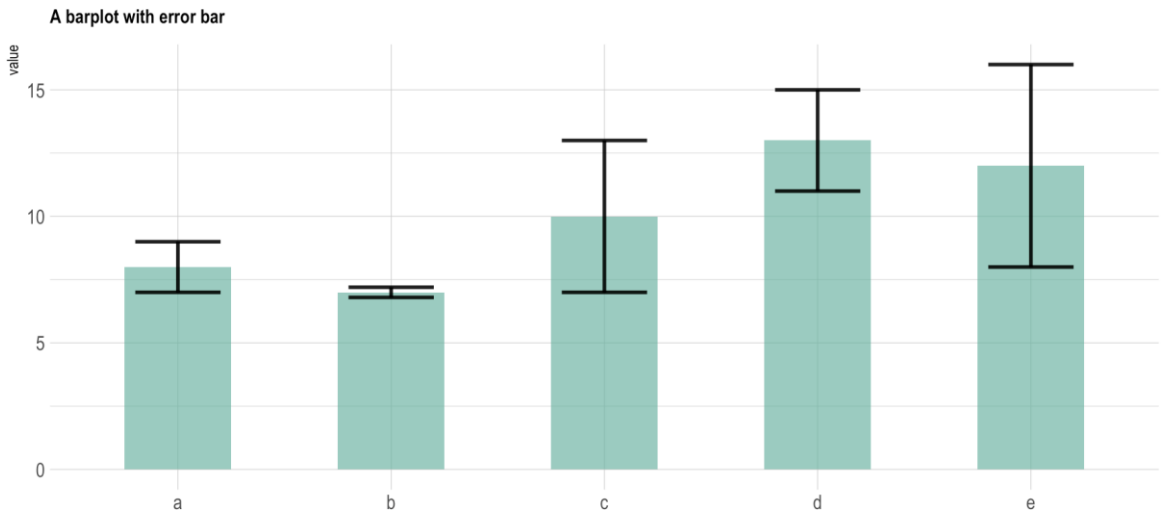
KORPORAAL, Michelle et FABRIKANT, Sara Irina. How are map-based decisions influenced by uncertainty visualization in risky and time-critical situations?. *Abstracts of the ICA*, 2019, vol. 1, p. 184.

KORPORAAL, Michelle, RUGINSKI, Ian T., et FABRIKANT, Sara Irina. Effects of uncertainty visualization on map-based decision making under time pressure. *Frontiers in Computer Science*, 2020, vol. 2, p. 32.

Assess a degree of uncertainty

Value with a confidence interval
Area with a probability value
of data presence or event
occurence

Visualizing uncertainty as a quantitative, or ordered, indicator

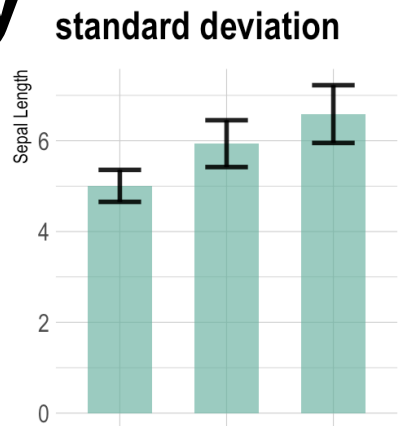


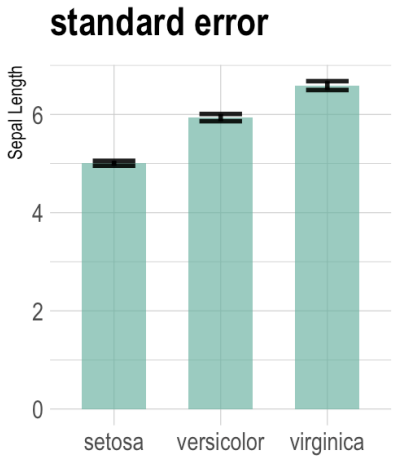
https://www.data-to-viz.com

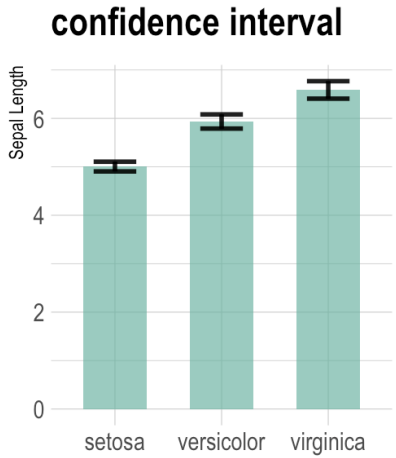
Assess a degree of uncertainty

Value with a confidence interval
Area with a probability value
of data presence or event
occurence

Visualizing uncertainty as a quantitative, or ordered, indicator





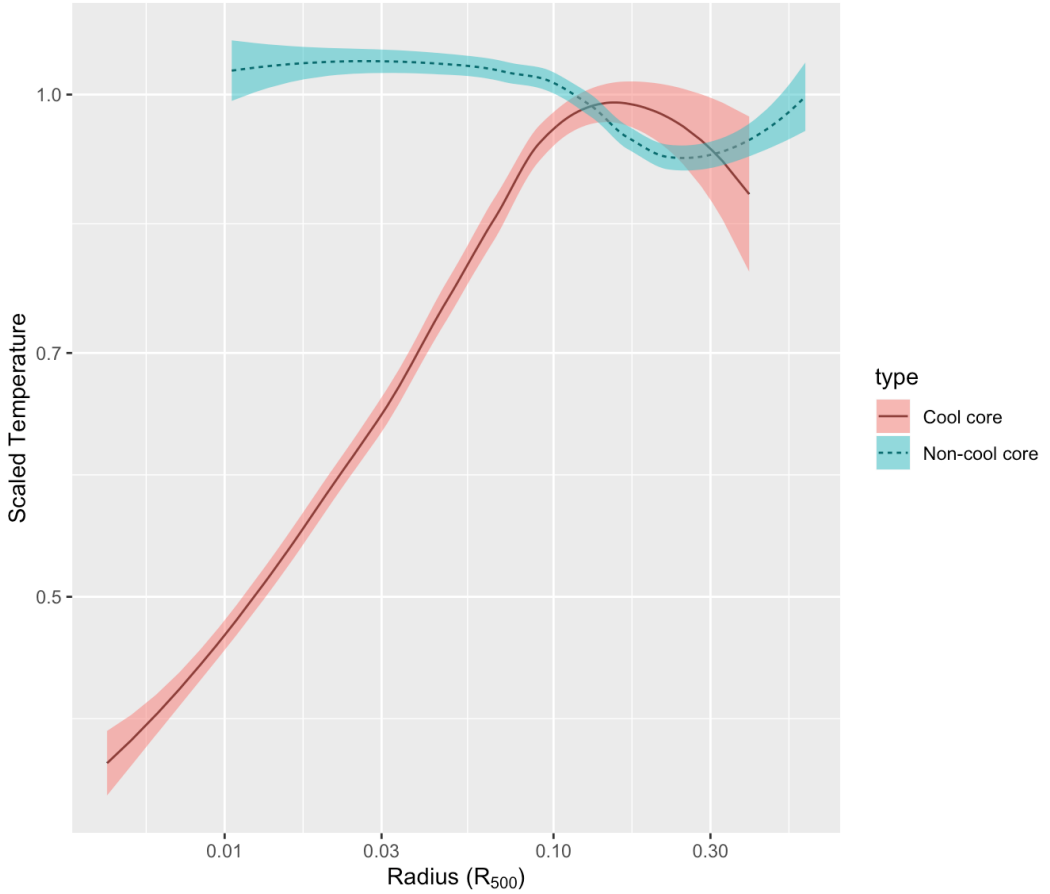


https://www.data-to-viz.com

Assess a degree of uncertainty

Value with a confidence interval
Area with a probability value of data presence or event occurence

Visualizing uncertainty as a quantitative, or ordered, indicator

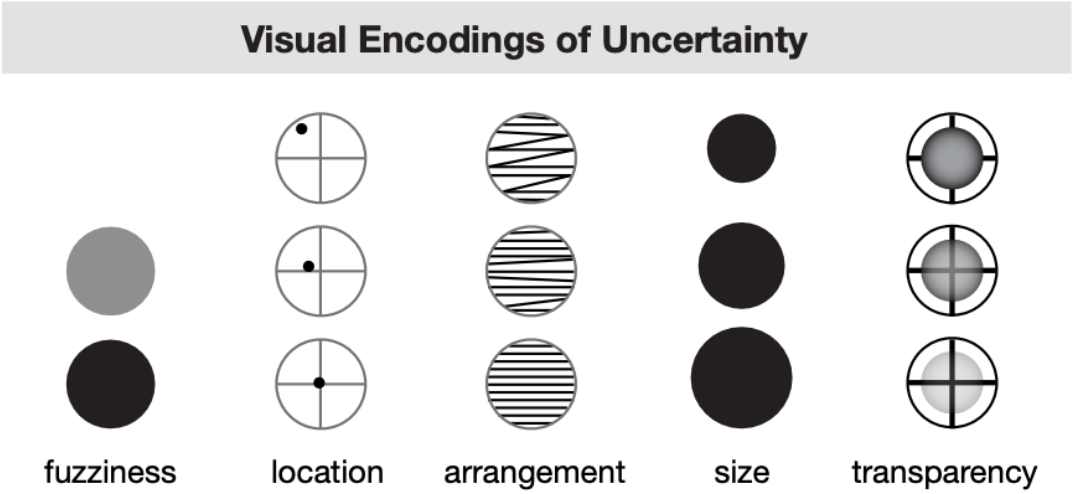


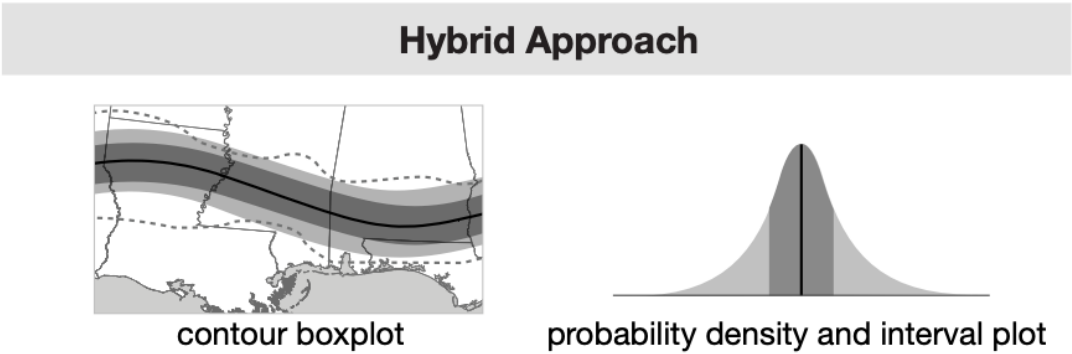
https://www.data-to-viz.com

Assess a degree of uncertainty

Value with a confidence interval
Area with a probability value of data presence or event occurence

Visualizing uncertainty as a quantitative, or ordered, indicator





Padilla, L., Kay, M. and Hullman, J. (2022). Uncertainty Visualization. In W. Piegorsch, R. Levine, H. Zhang, and T. Lee (Eds.), Computational Statistics in Data Science (pp. 405-421). Wiley.

Assess a degree of uncertainty

Visualizing uncertainty as a quantitative, or ordered, indicator



https://choucas.ign.fr/

Choucas project

Integration of heterogeneous data and spatial reasoning to assist in locating victims in mountainous terrain

Assess a degree of uncertainty

Visualizing uncertainty as a quantitative, or ordered, indicator



https://choucas.ign.fr/

Choucas project

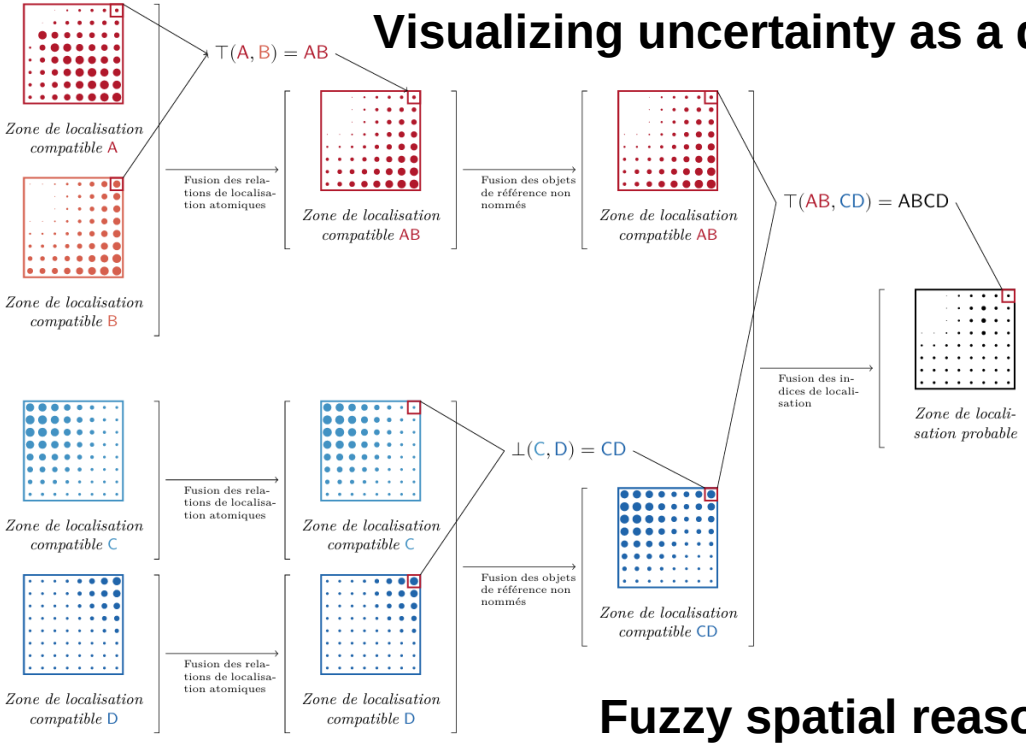
Integration of heterogeneous data and spatial reasoning to assist in locating victims in mountainous terrain

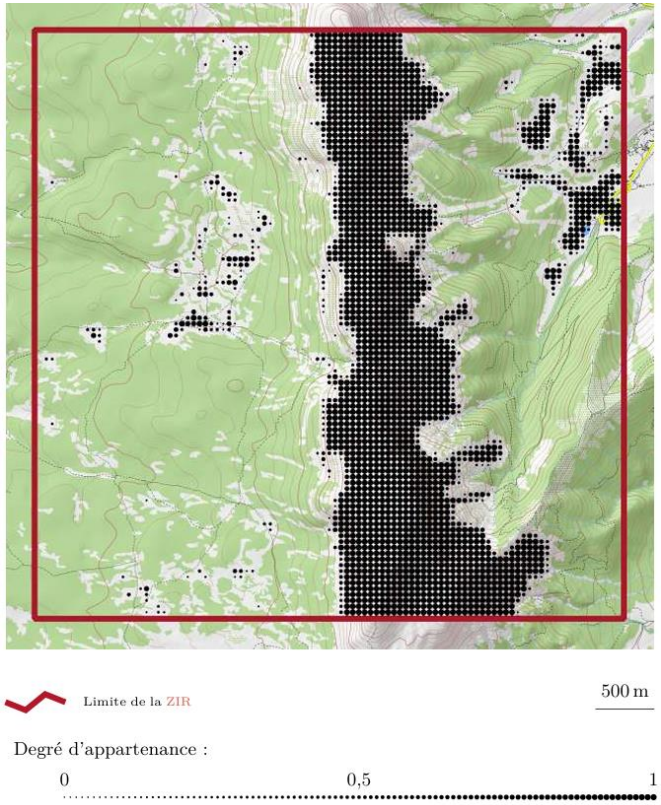
Somebody calls for help
« I set off toward Saint-Pierre from the Col de la Porte half an hour ago; I am near a forest and can see a river »

Where do we have to send the helicopter?

Assess a degree of uncertainty

Visualizing uncertainty as a quantitative, or ordered, indicator



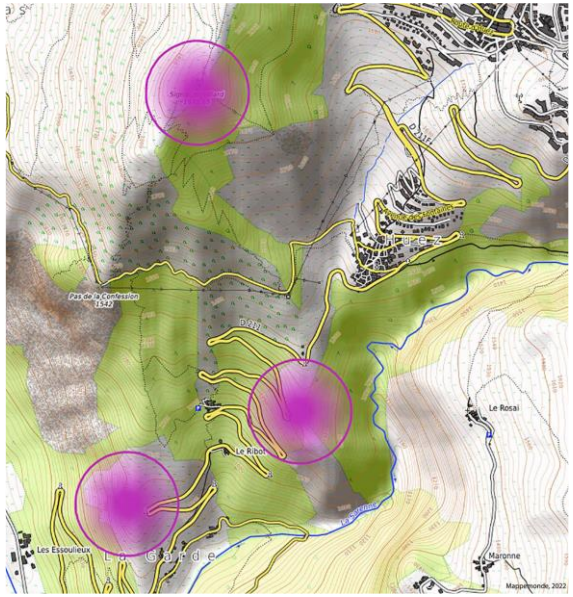


Fuzzy spatial reasoning

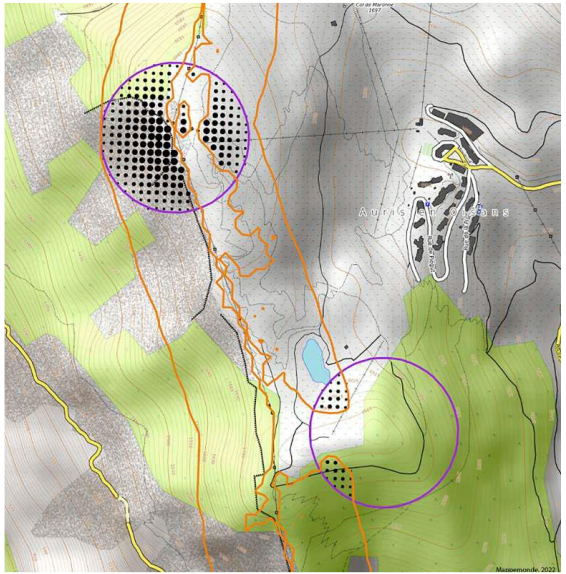
« The victim is not in a forest »

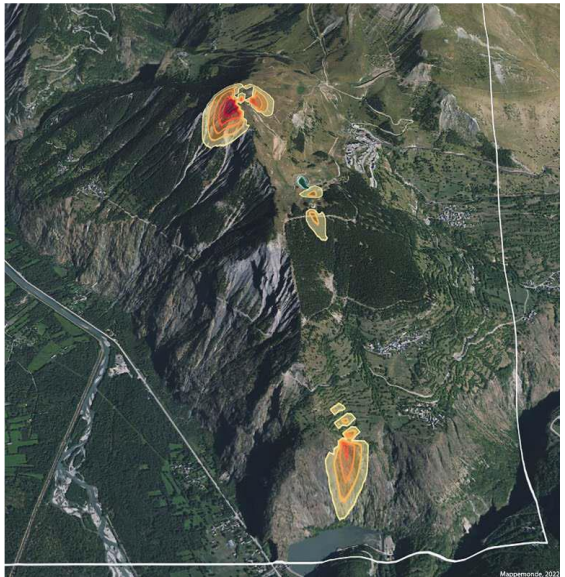
Bunel, M. (2023). Modélisation et raisonnement spatial flous pour l'aide à la localisation de victimes en montagne. *Cartes & géomatique*, (252), 93-95.

Assess a degree of uncertainty



Visualizing uncertainty as a quantitative, or ordered, indicator

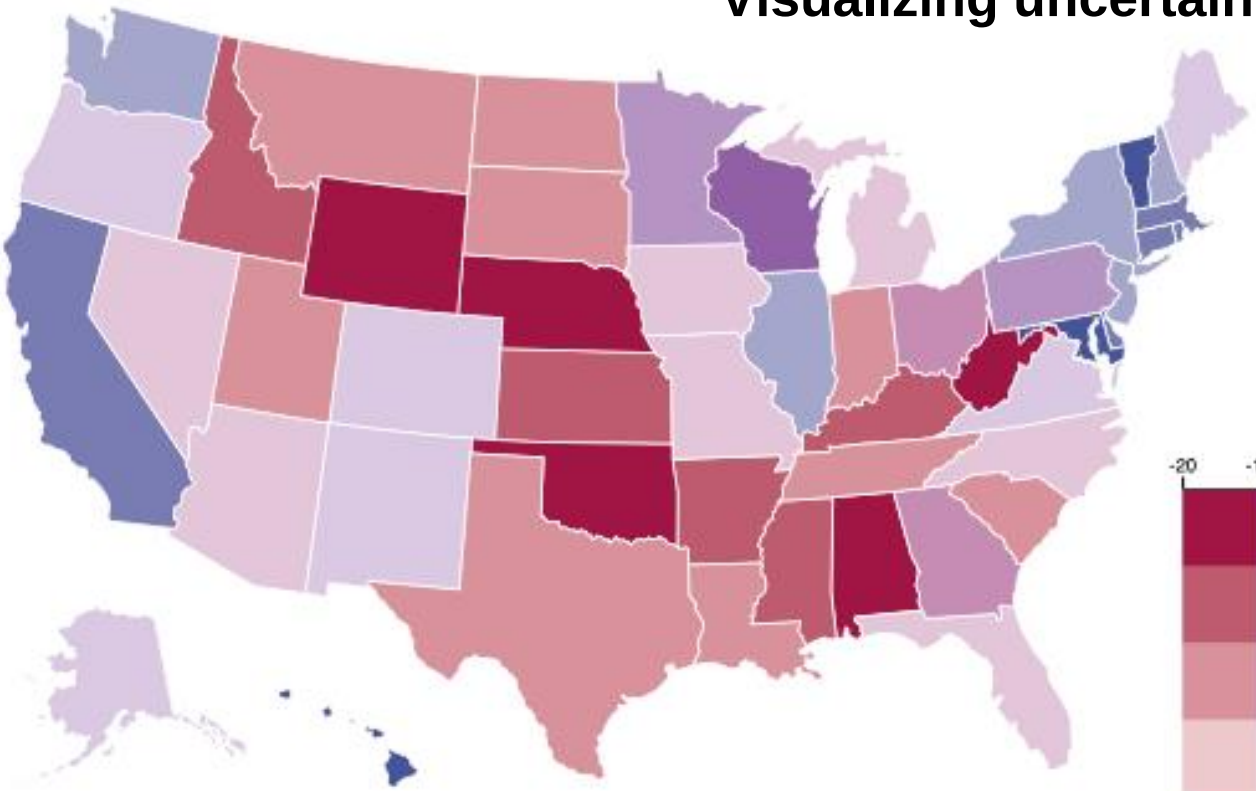


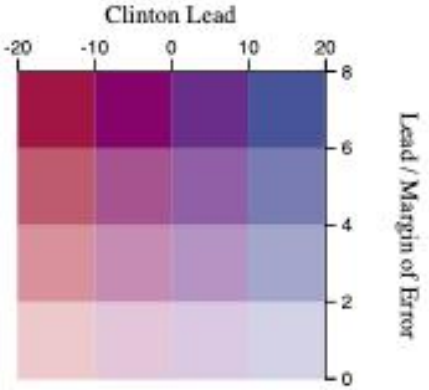


Viry, M. (2022). GASPAR—Geoviz Application for Searching And Rescuing People. *Mappemonde. Revue trimestrielle sur l'image géographique et les formes du territoire*, (134).

Assess a degree of uncertainty

Visualizing uncertainty as a quantitative, or ordered, indicator



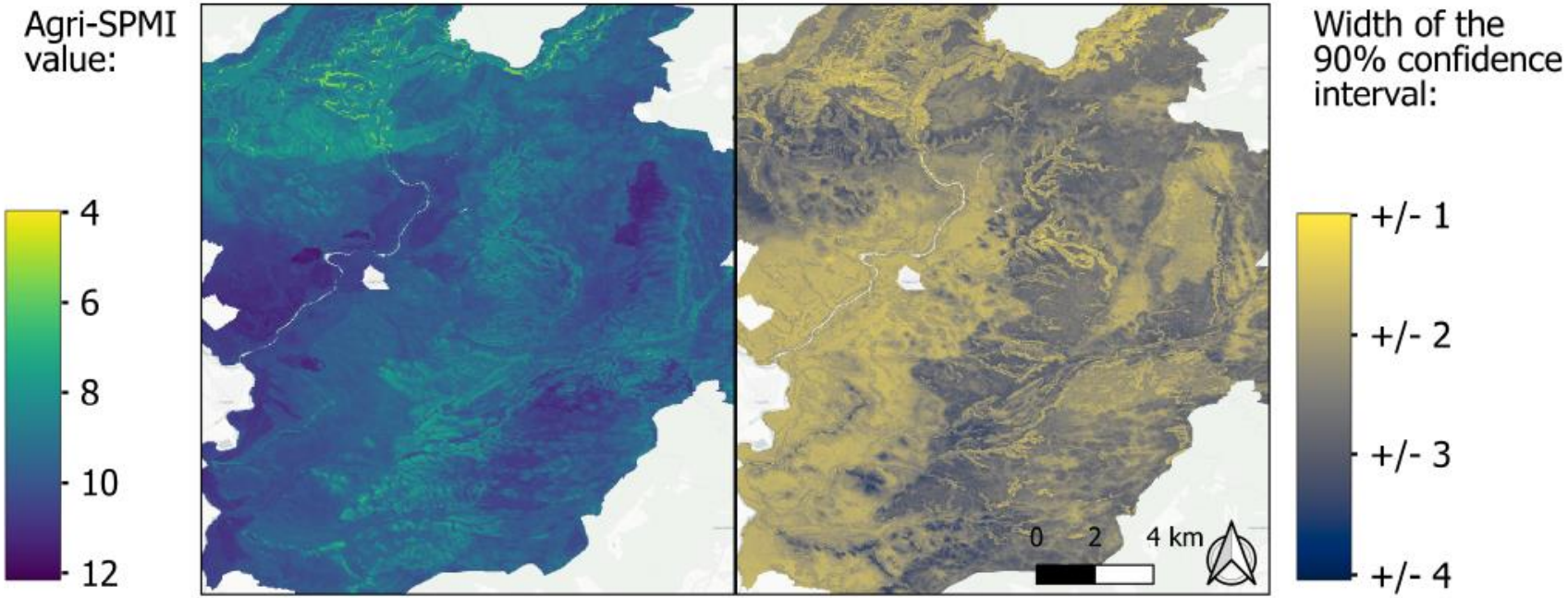


Correll, M., Moritz, D., & Heer, J. (2018, April). Value-suppressing uncertainty palettes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1-11).

Assess a degree of uncertainty

COURTEILLE, Léa, LAGACHERIE, Philippe, BOUKHELIFA, Nadia, et al. Using spatial aggregation of soil multifunctionality maps to support uncertainty-aware planning decisions. European Journal of Soil Science, 2024, vol. 75, no 4, p. e13523.

Visualizing uncertainty as a quantitative, or ordered, indicator



Assess a degree of uncertainty

Visualizing uncertainty as a quantitative, or ordered, indicator

Animation:

The further an object moves, the more uncertain the represented data is

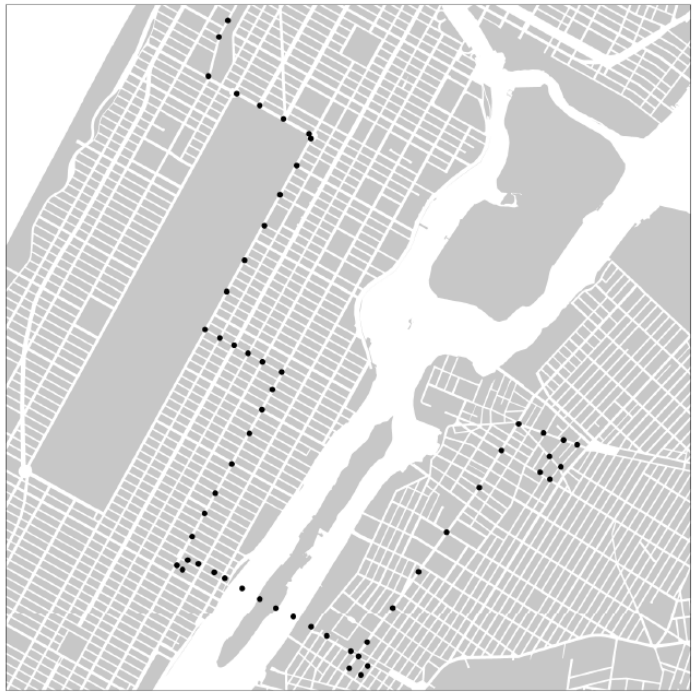


KEßLER, Carsten et LOTSTEIN, Enid. Animation as a visual indicator of positional uncertainty in geographic information. In : Geospatial Technologies for All: Selected Papers of the 21st AGILE Conference on Geographic Information Science 21. Springer International Publishing, 2018. p. 365-382.

Sound:

A different sound is play while moving the cursor on the map

LODHA, Suresh K., WILSON, Catherine M., et SHEEHAN, Robert E. LISTEN: sounding uncertainty visualization. In : *Proceedings of Seventh Annual IEEE Visualization'96*. IEEE, 1996. p. 189-195.



Analyzing different outcomes

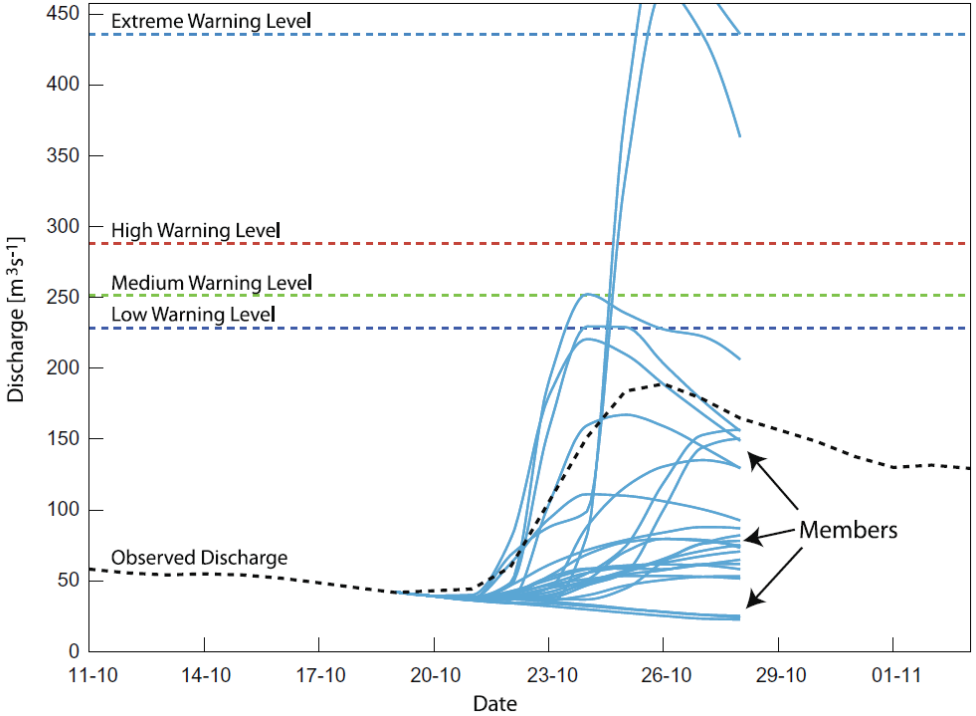
Probalistic approach

Quantify uncertainty by representing the distribution of the different possible outcomes

Analyzing different outcomes

Probalistic approach

Quantify uncertainty by representing the distribution of the different possible outcomes



Ensemble Plots

Ensemble simulation

Simulation of different scenarios (members)
≠ entry conditions
≠ models or model parameters
…analyzed as a all (ensemble)

Objectives: Identify
- *Most probable outcome* and its representativity
- *Possible outliers (black swan) scenarios*

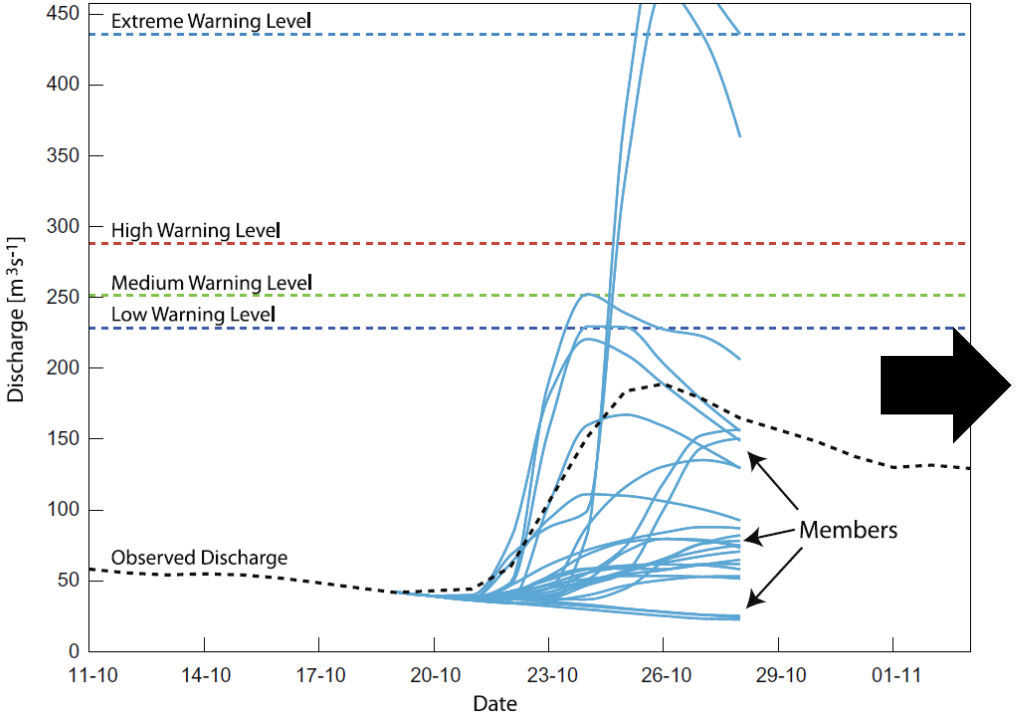
H.L. Cloke and F. Pappenberger, 2009, Ensemble flood forecasting: A review, Journal of Hydrology 375 (2009) 613–626

Potter, et al.. 2009. Ensemble-vis: A framework for the statistical visualization of ensemble data. In 2009 IEEE international conference on data mining workshops

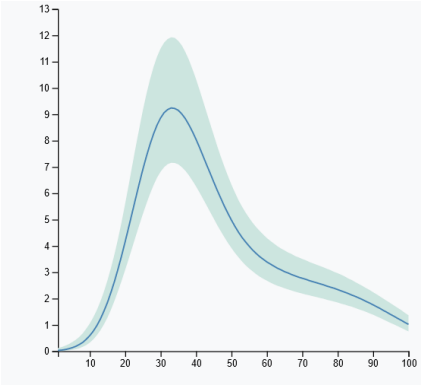
Analyzing different outcomes

Probalistic approach

Quantify uncertainty by representing the distribution of the different possible outcomes



Extraction of an ordered indicator
→ better readibility



H.L. Cloke and F. Pappenberger, 2009, Ensemble flood forecasting: A review, Journal of Hydrology 375 (2009) 613–626

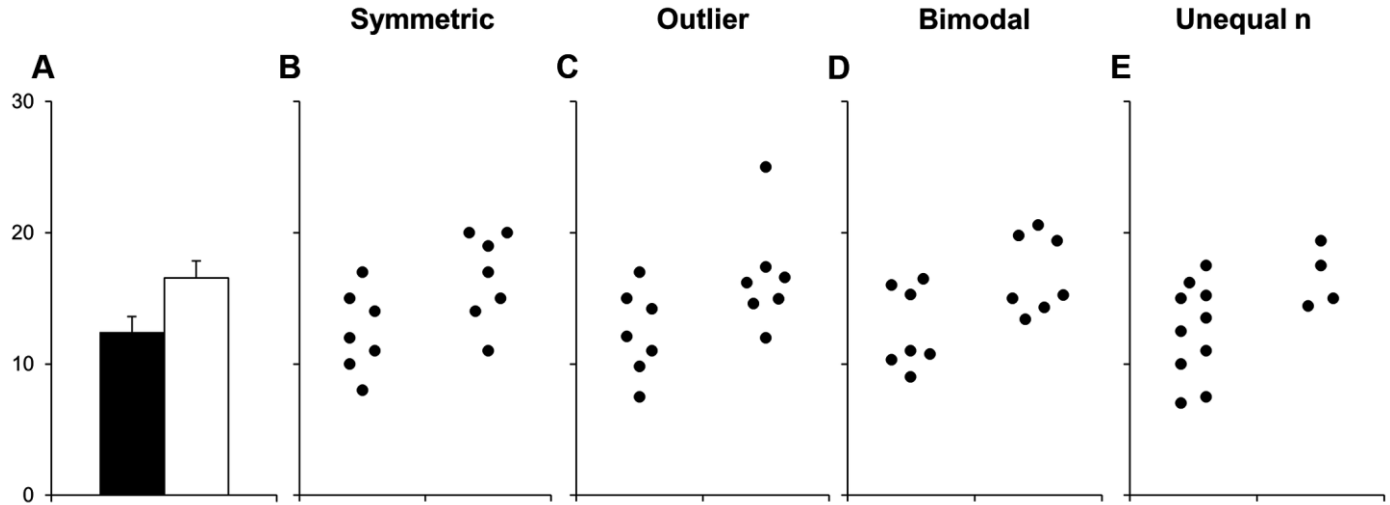
Potter, et al.. 2009. Ensemble-vis: A framework for the statistical visualization of ensemble data. In 2009 IEEE international conference on data mining workshops

Analyzing different outcomes

Probalistic approach

Quantify uncertainty by representing the distribution of the different possible outcomes

Extraction of an ordered indicator
→ better readibility



Weissgerber et al. 2015

Which indicator ?

How to quantify it ?

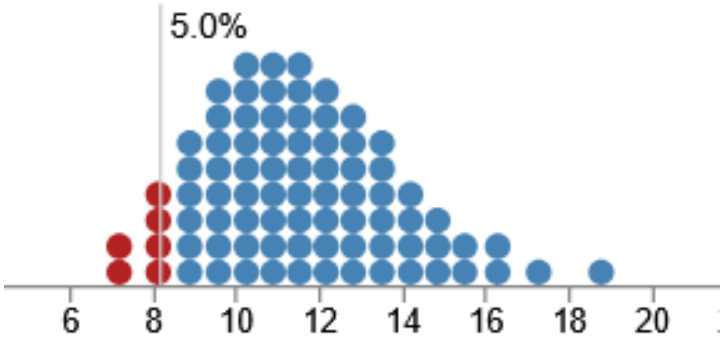
Possible lose of information

Analyzing different outcomes

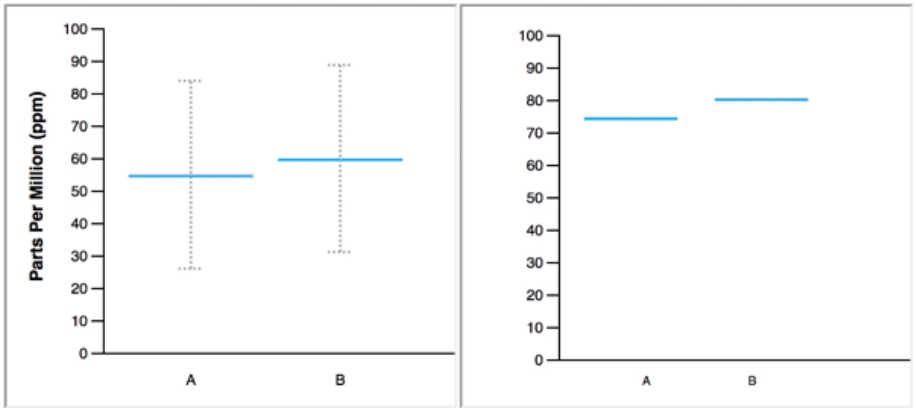
Probalistic approach

Quantify uncertainty by representing the distribution of the different possible outcomes

~~**Extraction of an ordered indicator**~~ **→ Directly visualizing the members distribution**



Quantile dot plots
https://vega.github.io



Hypothetical outcome plots (HOPs)
https://data.europa.eu

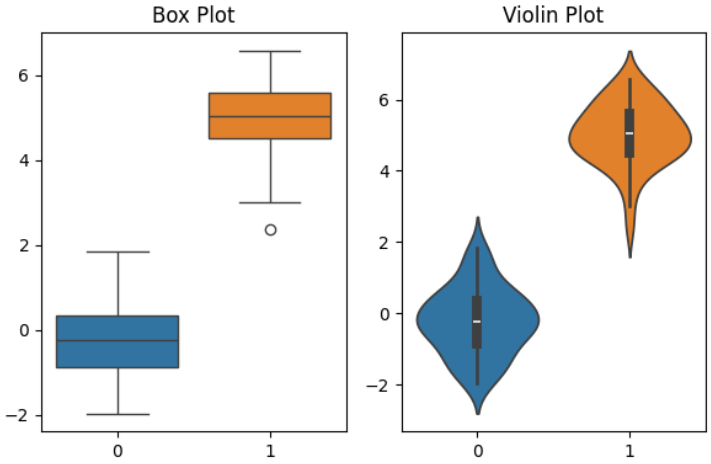
Padilla, L., Kay, M. and Hullman, J. (2022). Uncertainty Visualization. In W. Piegorsch, R. Levine, H. Zhang, and T. Lee (Eds.), Computational Statistics in Data Science (pp. 405-421). Wiley.

Analyzing different outcomes

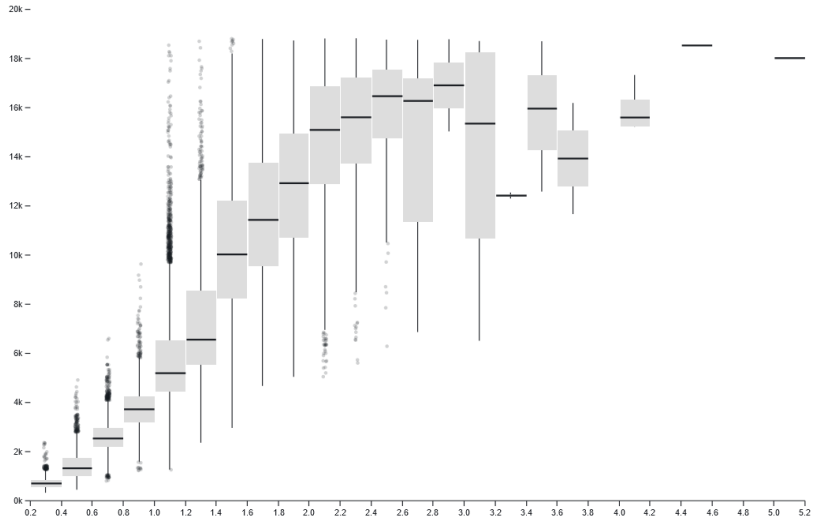
Probalistic approach

Quantify uncertainty by representing the distribution of the different possible outcomes

~~Extraction of an ordered indicator~~ → Visualizing several indicators to create more or less detailed representations of members distribution



Box plots ; Violin Plots
https://www.geeksforgeeks.org/



Padilla, L., Kay, M. and Hullman, J. (2022). Uncertainty Visualization. In W. Piegorsch, R. Levine, H. Zhang, and T. Lee (Eds.), Computational Statistics in Data Science (pp. 405-421). Wiley.

Analyzing different outcomes

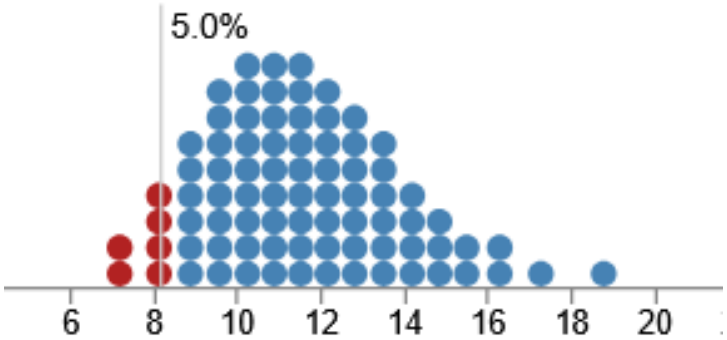
Probalistic approach

Quantify uncertainty by representing the distribution of the different possible outcomes

Visualize ensemble data relative to

- 0D data

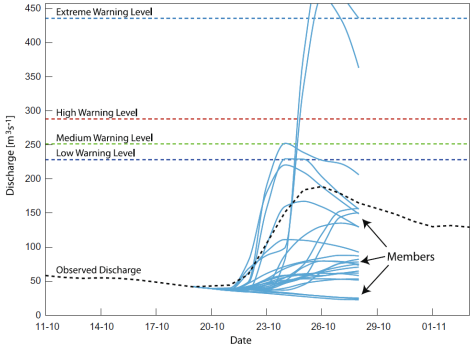
(different statistical values)



Is relatively easy.

- 1D data

(different time series)



Potter, et al.. 2009. Ensemble-vis: A framework for the statistical visualization of ensemble data. In 2009 IEEE international conference on data mining workshops

H.L. Cloke and F. Pappenberger, 2009, Ensemble flood forecasting: A review, Journal of Hydrology 375 (2009) 613–626

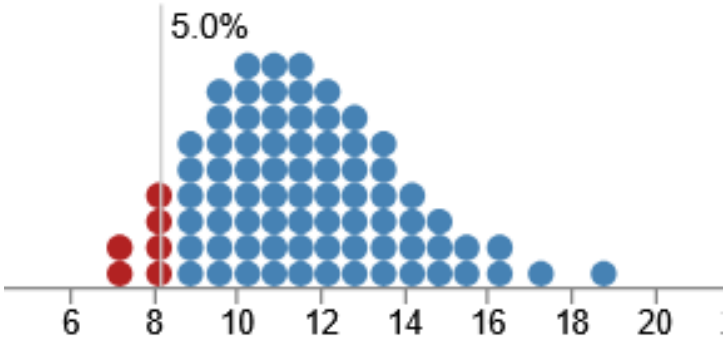
Analyzing different outcomes

Probalistic approach

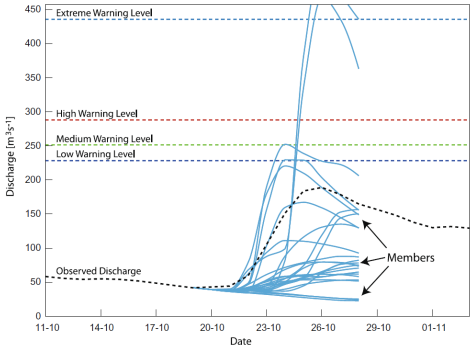
Quantify uncertainty by representing the distribution of the different possible outcomes

Visualize ensemble data relative to

- 0D data
(different statistical values)



- 1D data
(different time series)



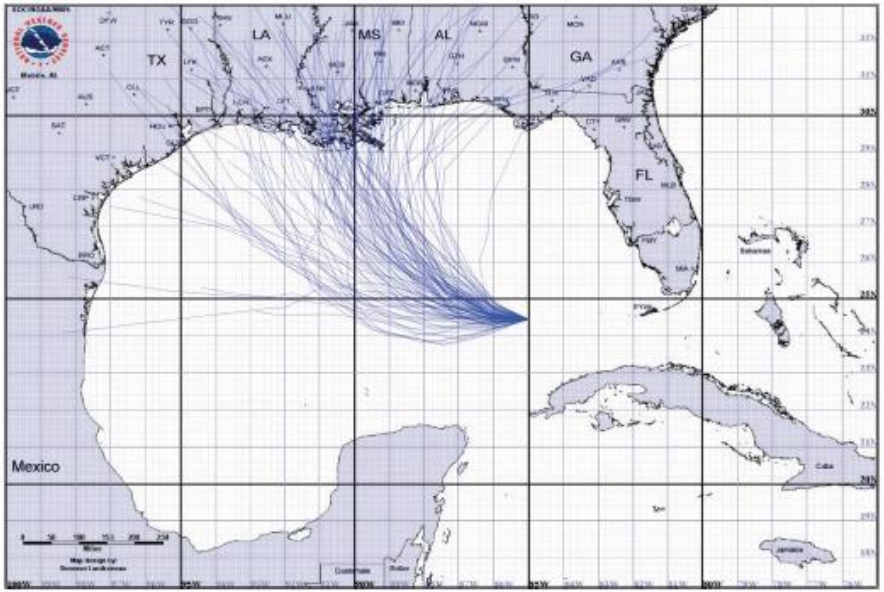
Potter, et al.. 2009. Ensemble-vis: A framework for the statistical visualization of ensemble data. In 2009 IEEE international conference on data mining workshops

H.L. Cloke and F. Pappenberger, 2009, Ensemble flood forecasting: A review, Journal of Hydrology 375 (2009) 613–626

Is relatively easy.

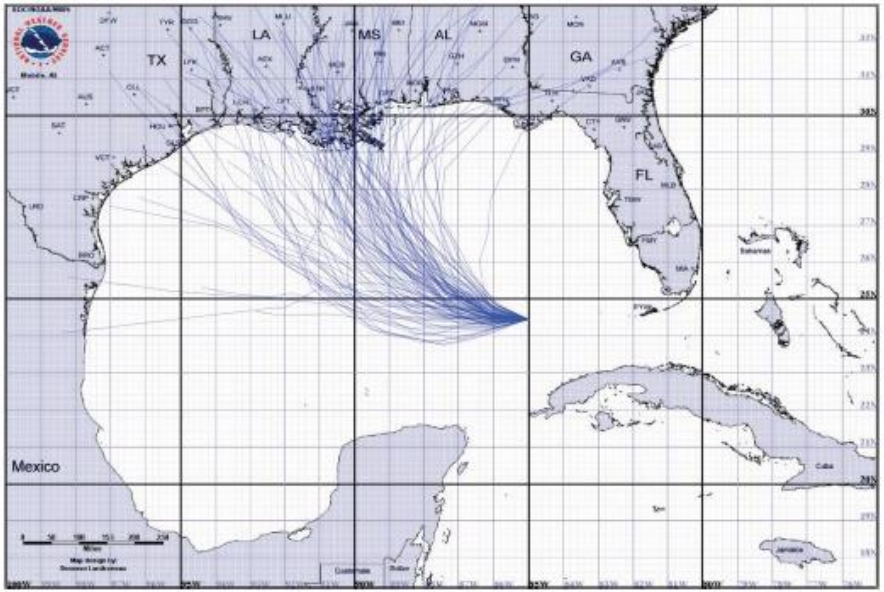
But what about ensemble data relative to 2D data (different spatial distribution)?

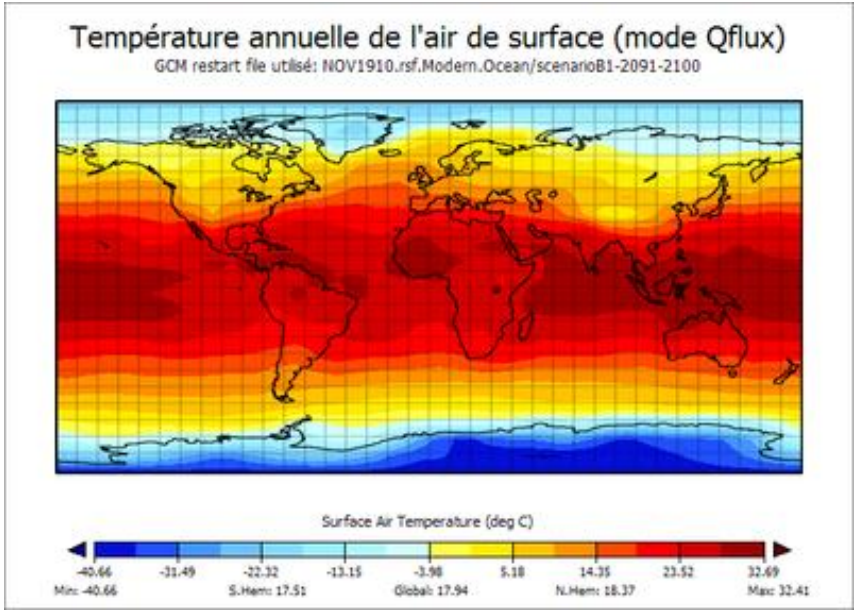
Analyzing different outcomes



Mirzargar, M., Whitaker, R. T., & Kirby, R. M. (2014). Curve boxplot: Generalization of boxplot for ensembles of curves. *IEEE transactions on visualization and computer graphics*, *20*(12), 2654-2663.

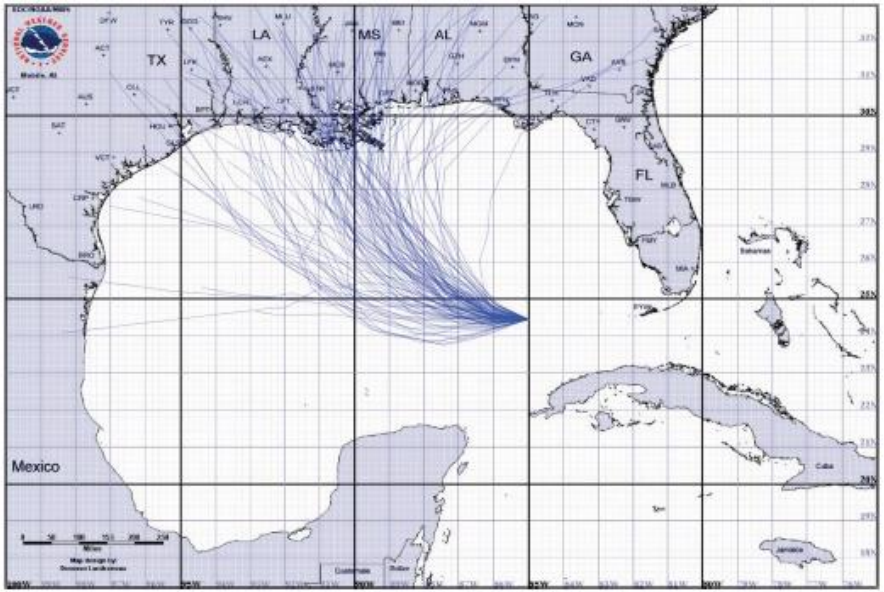
Analyzing different outcomes

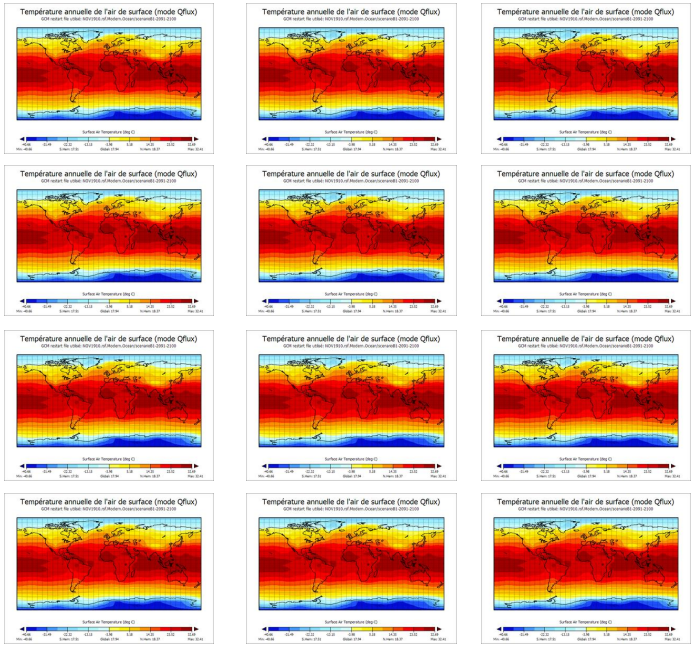




Mirzargar, M., Whitaker, R. T., & Kirby, R. M. (2014). Curve boxplot: Generalization of boxplot for ensembles of curves. *IEEE transactions on visualization and computer graphics*, *20*(12), 2654-2663.
https://acces.ens-lyon.fr

Analyzing different outcomes

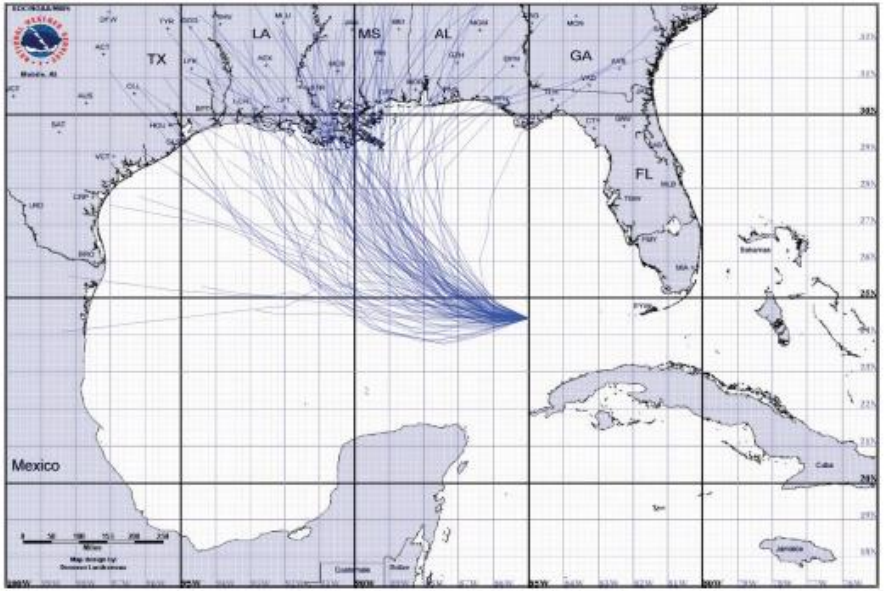




Mirzargar, M., Whitaker, R. T., & Kirby, R. M. (2014). Curve boxplot: Generalization of boxplot for ensembles of curves. *IEEE transactions on visualization and computer graphics*, *20*(12), 2654-2663.
https://acces.ens-lyon.fr

Analyzing different outcomes

A example *of postage stamps showing PMSL forecasts from ensemble control, and all 50 ensemble members. Source:* https://confluence.ecmwf.int/display/FUG/Section+8.1.1+Basic+ensemble+products#Section8.1.1Basicensembleproducts-Postagestamps

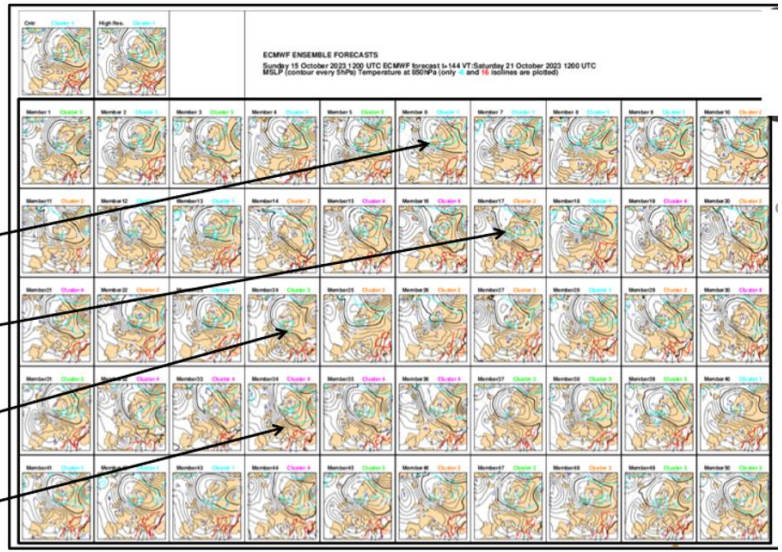


Member No6 representative of Cluster No1

Member No17 representative of Cluster No2

Member No24 representative of Cluster No3

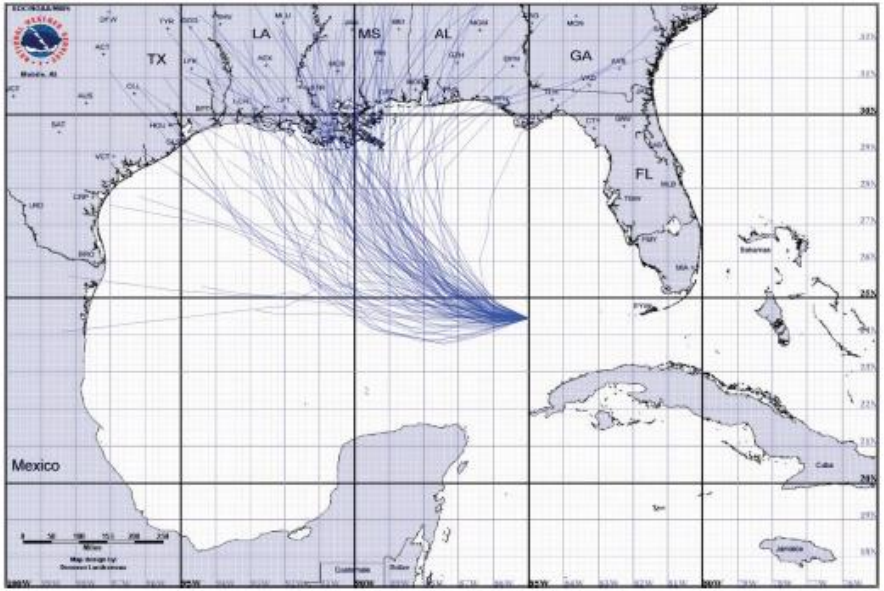
Member No34 representative of Cluster No4



Mirzargar, M., Whitaker, R. T., & Kirby, R. M. (2014). Curve boxplot: Generalization of boxplot for ensembles of curves. *IEEE transactions on visualization and computer graphics*, *20*(12), 2654-2663.
https://acces.ens-lyon.fr

Analyzing different outcomes

A example *of postage stamps showing PMSL forecasts from ensemble control, and all 50 ensemble members. Source:* https://confluence.ecmwf.int/display/FUG/Section+8.1.1+Basic+ensemble+products#Section8.1.1Basicensembleproducts-Postagestamps

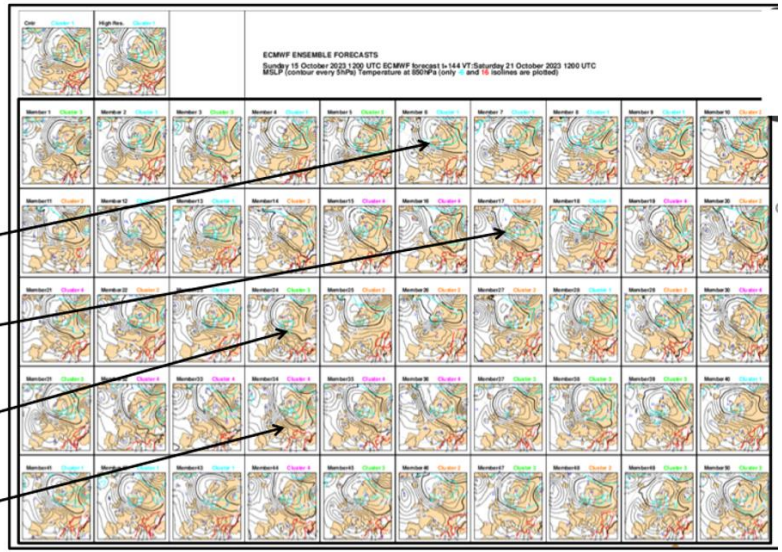


Member No6 representative of Cluster No1

Member No17 representative of Cluster No2

Member No24 representative of Cluster No3

Member No34 representative of Cluster No4



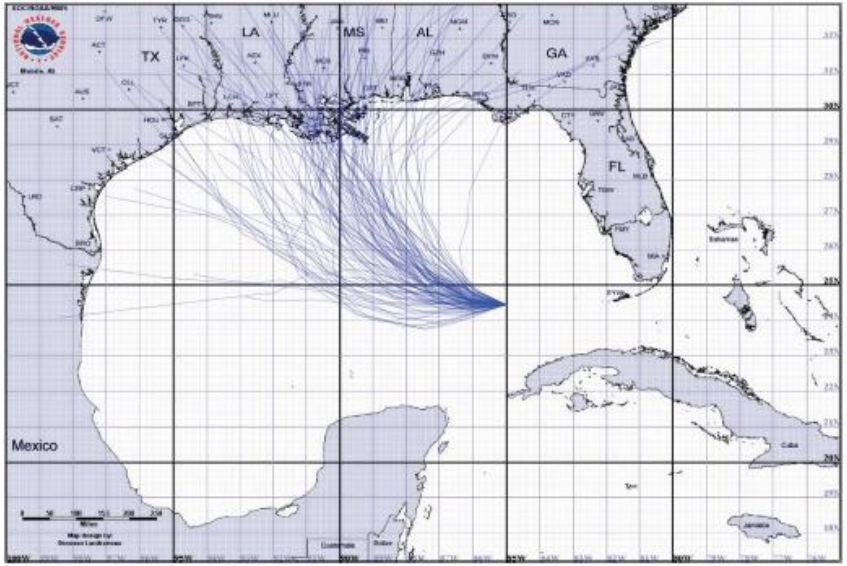
Traditional cartographic representation approaches are difficult to apply to ensemble data. (Wang, 2015)

Mirzargar, M., Whitaker, R. T., & Kirby, R. M. (2014). Curve boxplot: Generalization of boxplot for ensembles of curves. *IEEE transactions on visualization and computer graphics*, *20*(12), 2654-2663.
https://acces.ens-lyon.fr

Analyzing different outcomes

In most cases, you have to go through

- Aggregation (calculating indicators summarizing ensemble)
- Composition (co-visualizing several indicators through composed views or symbols)

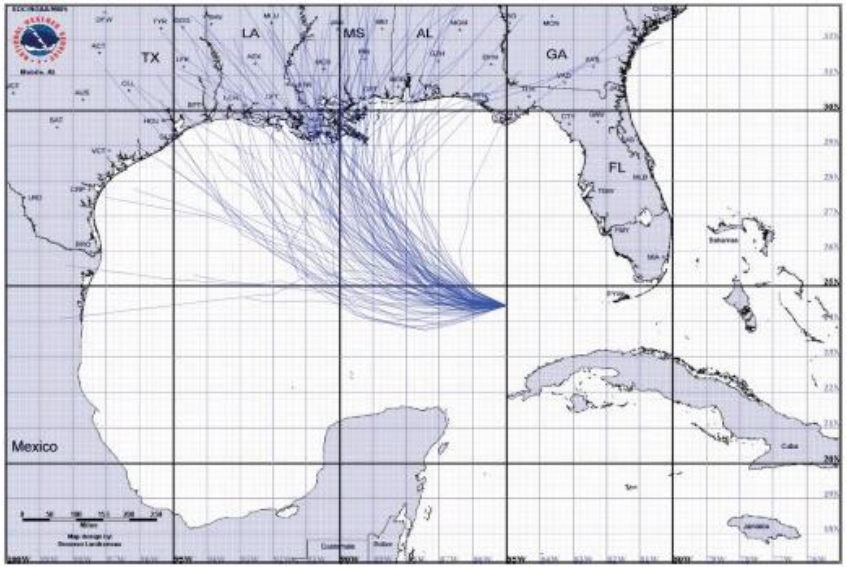


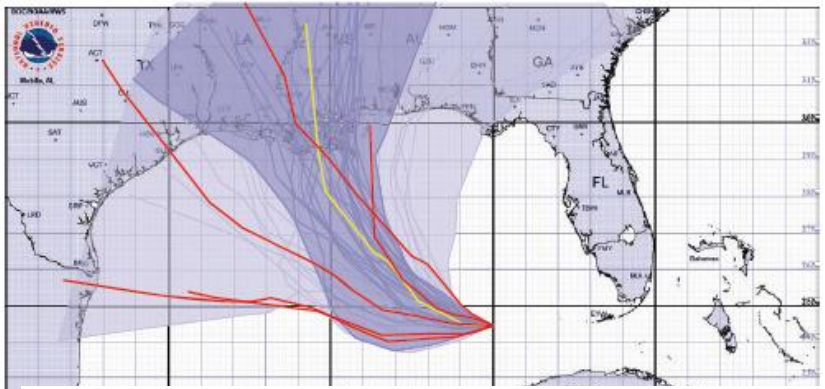
Mirzargar, M., Whitaker, R. T., & Kirby, R. M. (2014). Curve boxplot: Generalization of boxplot for ensembles of curves. *IEEE transactions on visualization and computer graphics*, *20*(12), 2654-2663.

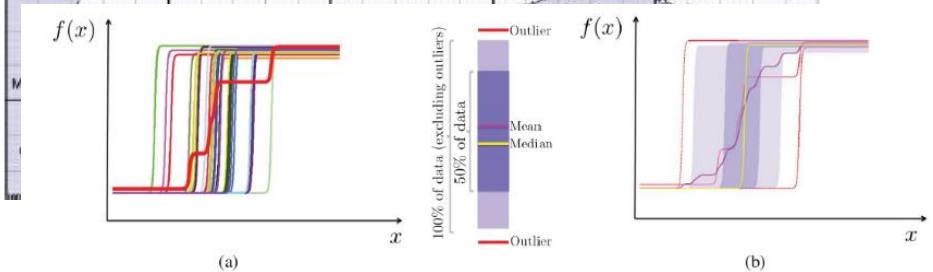
Analyzing different outcomes

In most cases, you have to go through

- Aggregation (calculating indicators summarizing ensemble)
- Composition (co-visualizing several indicators through composed views or symbols)





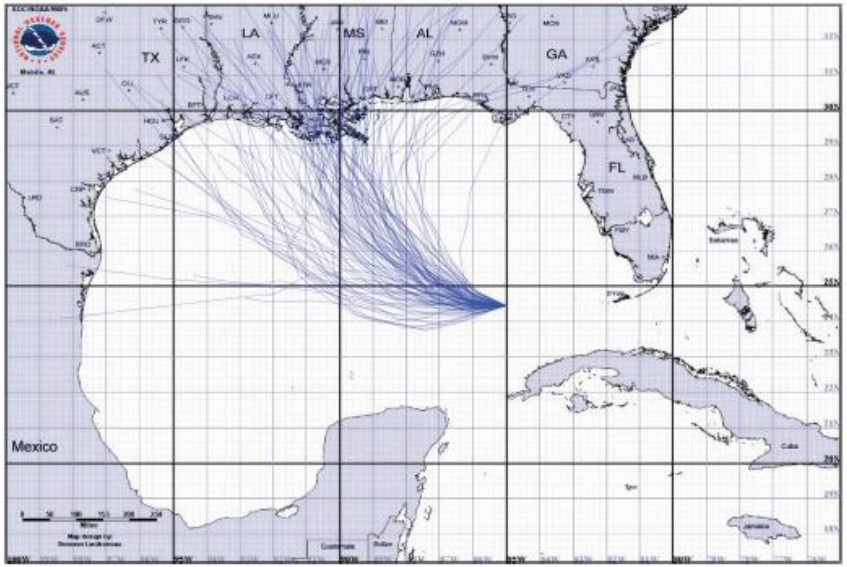


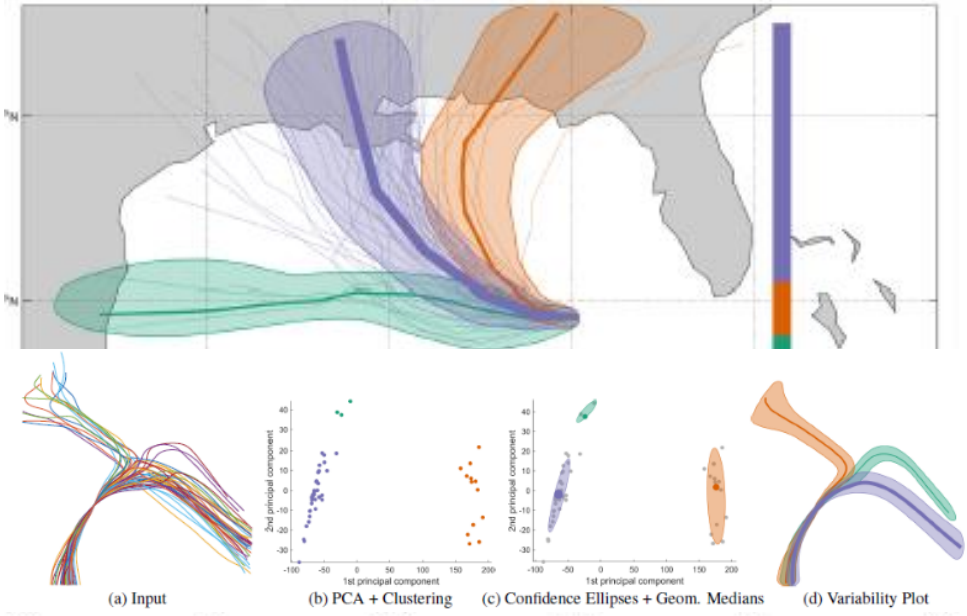
Mirzargar, M., Whitaker, R. T., & Kirby, R. M. (2014). Curve boxplot: Generalization of boxplot for ensembles of curves. *IEEE transactions on visualization and computer graphics*, *20*(12), 2654-2663.

Analyzing different outcomes

In most cases, you have to go through

- Aggregation (calculating indicators summarizing ensemble)
- Composition (co-visualizing several indicators through composed views or symbols)





Ferstl, F., Bürger, K., & Westermann, R. (2015). Streamline variability plots for characterizing the uncertainty in vector field ensembles. IEEE Transactions on Visualization and Computer Graphics, 22(1), 767-776.

Analyzing different outcomes

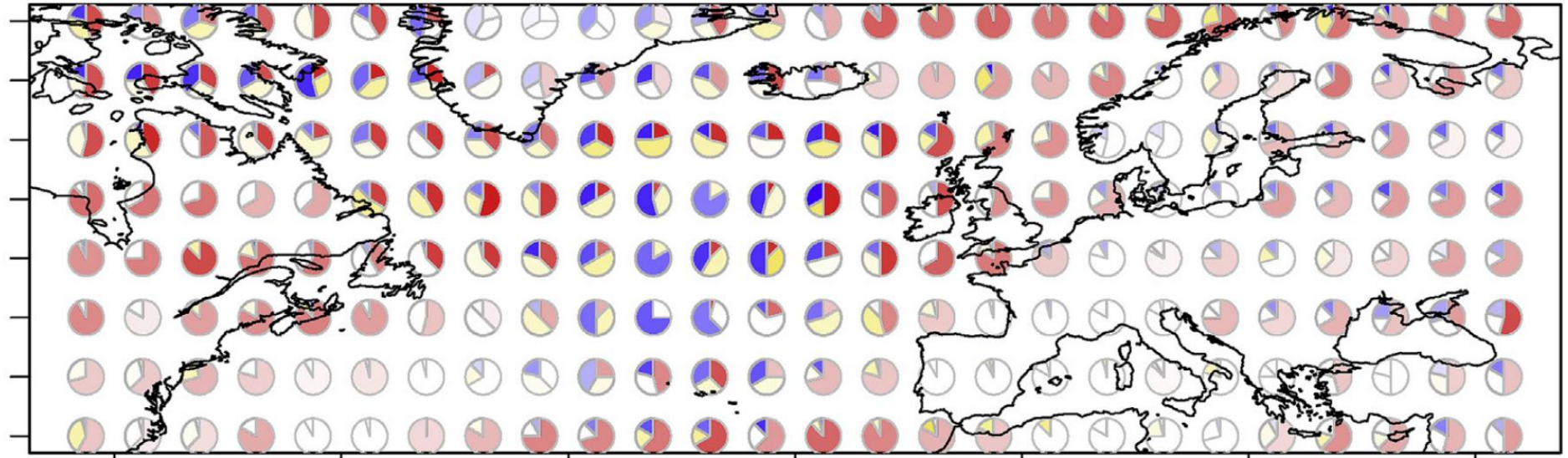
In most cases, you have to go through

- Aggregation (calculating indicators summarizing ensemble)
- Composition (co-visualizing several indicators through composed views or symbols)

2–meter air temperature, Dec to Feb, 2016

Reference data: NCEP; Hindcast: CFS (24 members); 1983–2010

An R package to visualize and communicate uncertainty in seasonal climate prediction M.D. Frías a, * , M. Iturbide b , R. Manzanas b , J. Bedia a , J. Fernandez a , S. Herrera a , A.S. Cofino~ a , J.M. Gutierrez b, Environmental Modelling & Software 99 (2018) 101e110



Analyzing different outcomes

In most cases, you have to go through

- Aggregation (calculating indicators summarizing ensemble)
- Composition (co-visualizing several indicators through composed views or symbols)

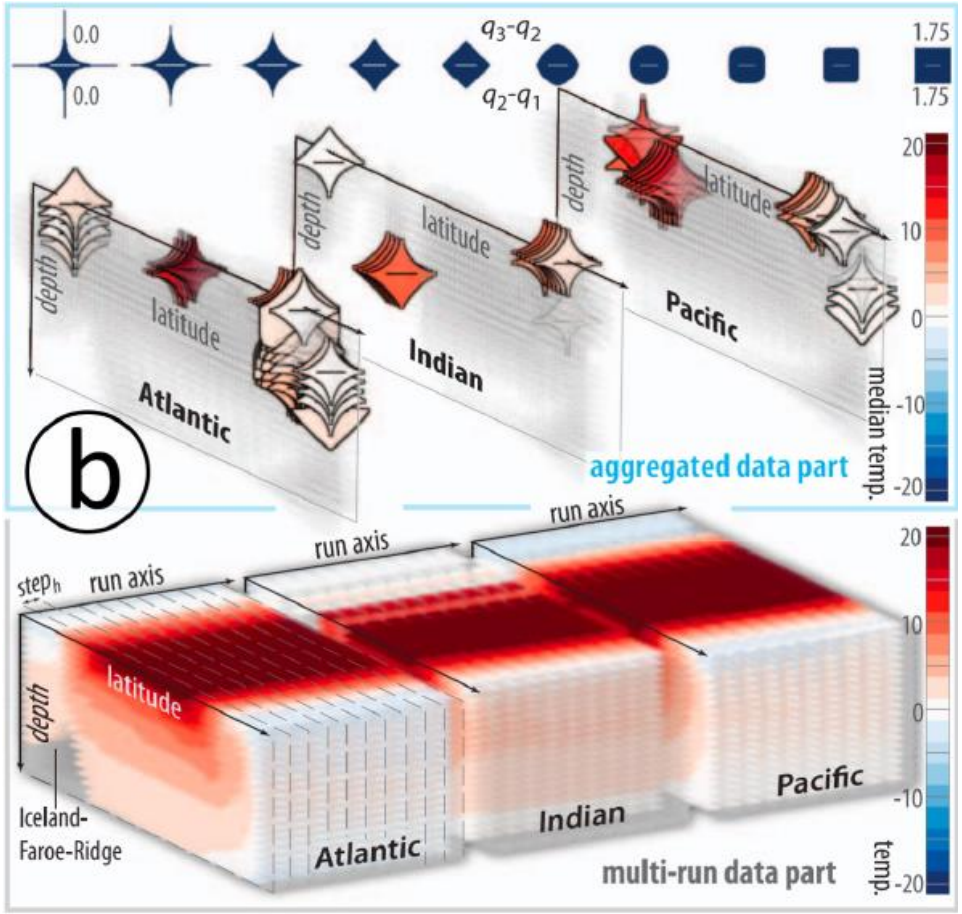


image courtesy of Kehrer et al. [62] © 2011 IEEE.
Visualization and Visual Analysis of Ensemble Data: A Survey
Junpeng Wang, Subhashis Hazarika, Cheng Li, and Han-Wei Shen

gridviz eurostat

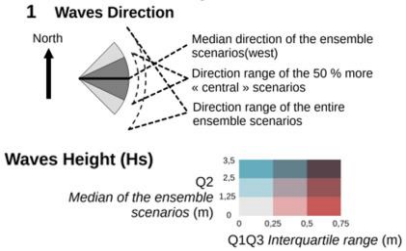


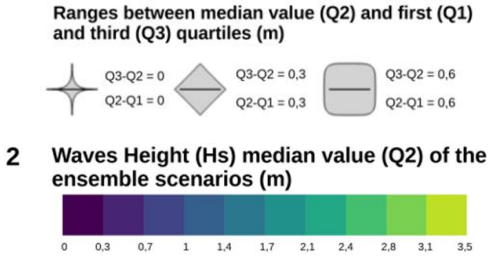


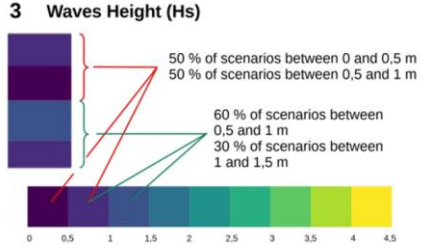


GLYPH

LEGEND







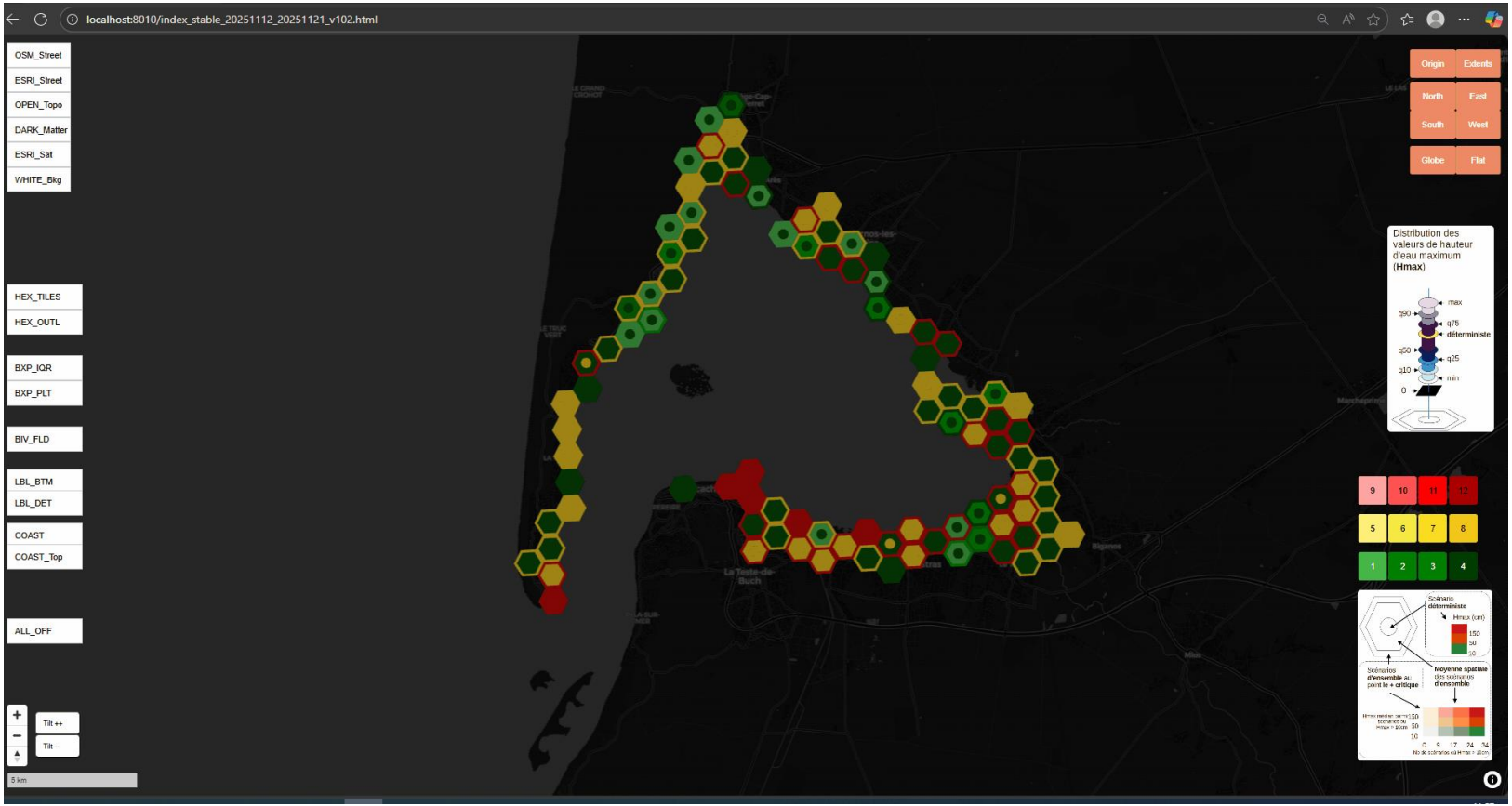
oracles

Visualizing uncertainty from meteo-oceanic ensemble data

Julius M. Bañgate, Jacques Gautier, Sidonie Christophe, Déborah Idier, Denis Paradis, Sophie Lecacheux

VIS2025

IEEE Workshop on Uncertainty Visualization: Unraveling Relationships of Uncertainty, AI, and Decision-Making
in conjunction with IEEE VIS 2025, Vienna, Austria, Nov. 2 – 7, 2025



oracles

2D and 3D visualisation of ensemble coastal flood forecasts for decision making under uncertainty.

Julius M. Bañgate, Jacques Gautier, Sidonie Christophe, Déborah Idier, Denis Paradis, Sophie Lecacheux

Visualizing data uncertainty

Visualizing the very concept of uncertainty

Discriminate uncertain data from the others

Assess a degree of uncertainty

Analyzing different outcomes

Visualizing data uncertainty

Visualizing the very concept of uncertainty

Discriminate uncertain data from the others

Assess a degree of uncertainty

Analyzing different outcomes

User expertise

Visualizing data uncertainty

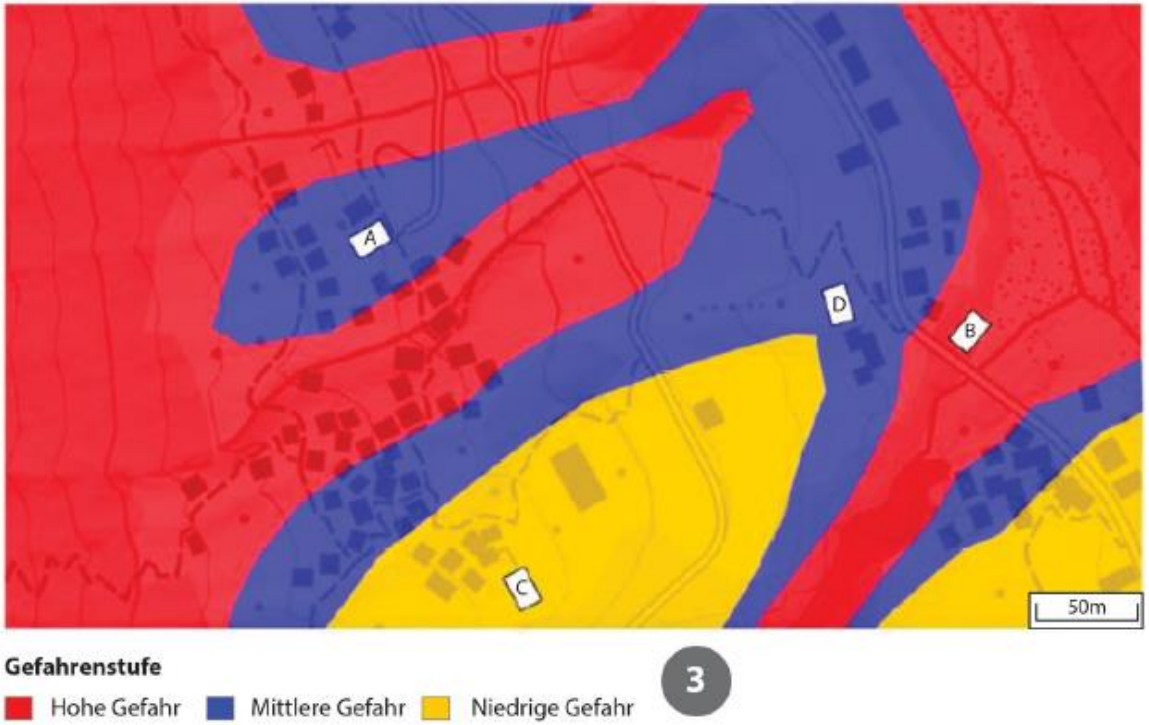
A domain still under exploration: a lot is still to be investigated…

…especially how users interpret uncertainty

Visualizing data uncertainty

A domain still under exploration: a lot is still to be investigated…

…especially how users interpret uncertainty



KÜBLER, Isabella, RICHTER, Kai-Florian, et FABRIKANT, Sara Irina. Against all odds: multicriteria decision making with hazard prediction maps depicting uncertainty. *Annals of the American Association of Geographers*, 2020, vol. 110, no 3, p. 661-683.

Visualizing data uncertainty

A domain still under exploration: a lot is still to be investigated…



…especially how users interpret uncertainty

« Uncertainty at the class boundaries is understood as such by our participants »

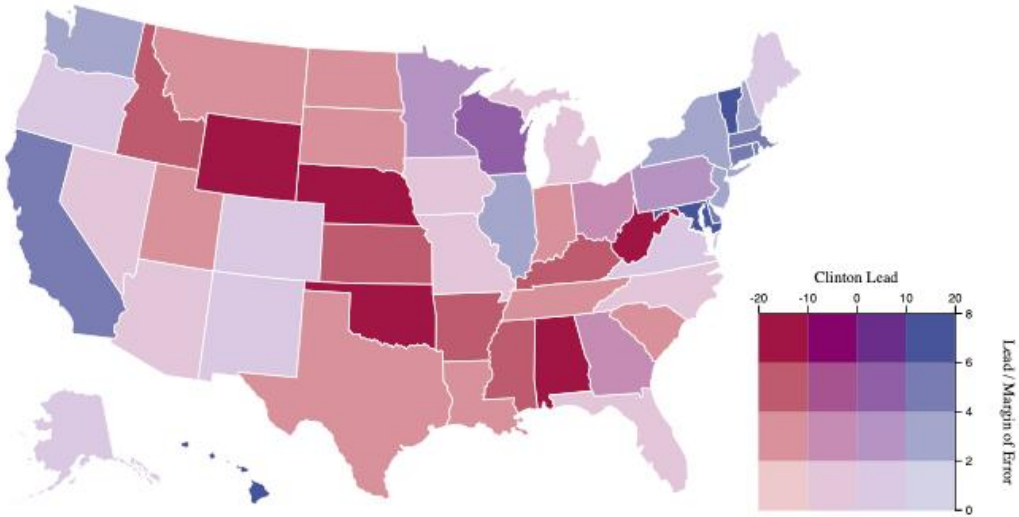
« Using visualizations with uncertainty led them instead to choose more houses in risk-prone areas than using visualizations without uncertainty depiction. »

KÜBLER, Isabella, RICHTER, Kai-Florian, et FABRIKANT, Sara Irina. Against all odds: multicriteria decision making with hazard prediction maps depicting uncertainty. *Annals of the American Association of Geographers*, 2020, vol. 110, no 3, p. 661-683.

Visualizing data uncertainty

A domain still under exploration: a lot is still to be investigated…

…especially how users interpret uncertainty

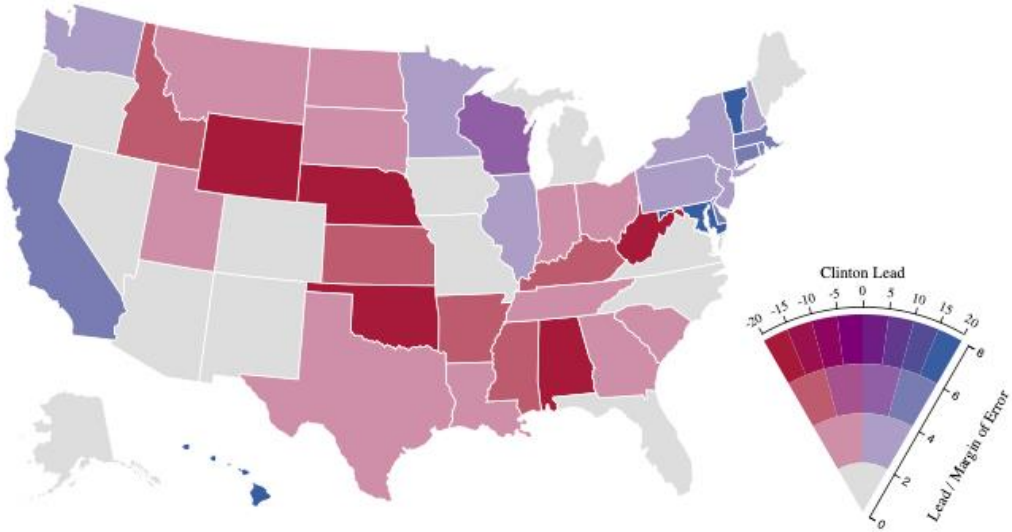


Correll, M., Moritz, D., & Heer, J. (2018, April). Value-suppressing uncertainty palettes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1-11).

Visualizing data uncertainty

A domain still under exploration: a lot is still to be investigated…

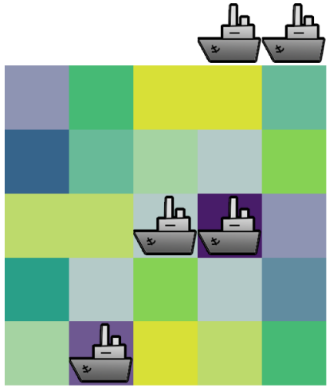
…especially how users interpret uncertainty

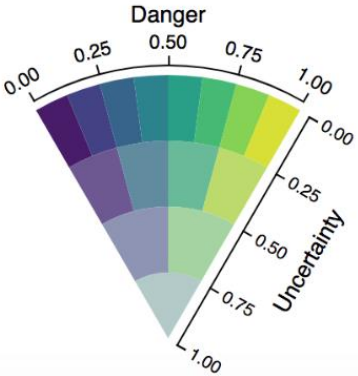


Correll, M., Moritz, D., & Heer, J. (2018, April). Value-suppressing uncertainty palettes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1-11).

Visualizing data uncertainty

A domain still under exploration: a lot is still to be investigated…





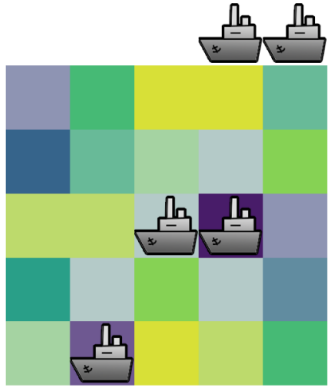
…especially how users interpret uncertainty

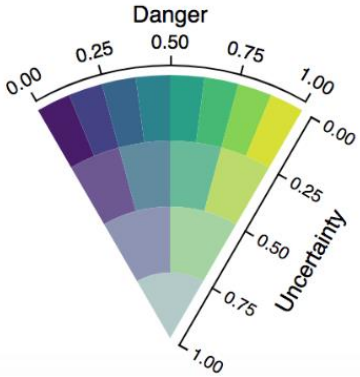
Correll, M., Moritz, D., & Heer, J. (2018, April). Value-suppressing uncertainty palettes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1-11).

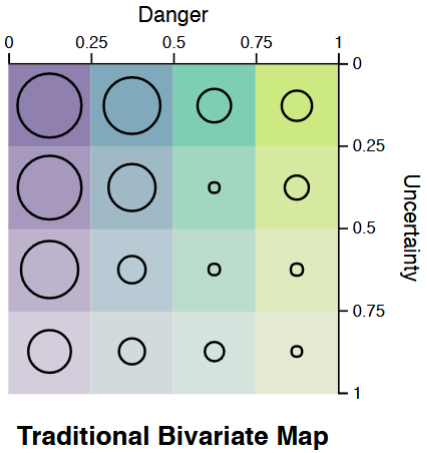
Visualizing data uncertainty

A domain still under exploration: a lot is still to be investigated…

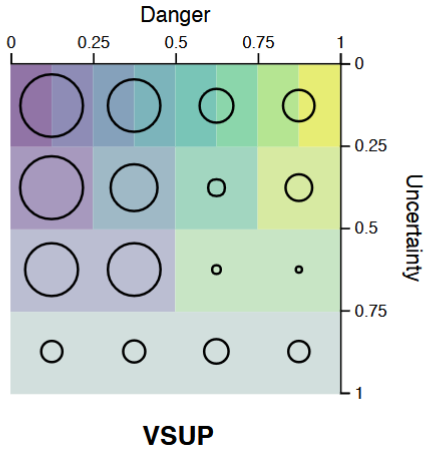
…especially how users interpret uncertainty



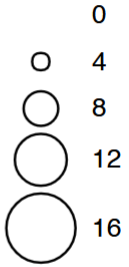




Traditional Bivariate Map



VSUP



Correll, M., Moritz, D., & Heer, J. (2018, April). Value-suppressing uncertainty palettes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1-11).

Visualizing data uncertainty

A domain still under exploration: a lot is still to be investigated…

…especially how users interpret uncertainty

Can be dependent on the how uncertainty is depicted, but also on the user and the task, which could promote risk averse or risk seeking decision making

According to the context, some designs can promote on or the other